

BLOCK 4

EXCHANGE RATE AND BALANCE OF PAYMENTS

Unit 7	Exchange Rate Regimes
Unit 8	Components of Balance of Payments
Unit 9	Impossible Trinity: Alternative Scenarios
Unit 10	Approaches to Balance of Payments

BLOCK INTRODUCTION

This block comprises of four units.

Unit 7: **Exchange Rate Regimes** deliberates on the concepts of foreign exchange, exchange rate, and exchange rate regimes. It distinguishes between different forms of exchange rate regimes: fixed, floating, and managed float along with their advantages and disadvantages. To delve deeper into the various exchange rate regimes, the unit traces briefly the historical development of international exchange rate regimes and examine operational aspects of the current exchange rate regimes. It also provides an overview of India's current exchange rate regime.

Unit 8: **Components of Balance of Payments** examines the framework used to summarize a country's international transactions: Balance of Payments (BOP). While explaining the concept of BOP, the Unit describes the components of the current and capital account of BOP and highlights the significance of autonomous and accommodating items. The unit also provides comparison between BOP and BOT, current and capital account of BOP and autonomous and accommodating transactions of BoP. The Unit also throws light on nature and implications of BOP disequilibrium. Along with discussing the issues associated with BOP disequilibrium, this Unit also suggests policy measures for correcting disequilibrium.

Unit 9: **Impossible Trinity: Alternative Scenarios** elucidates Macroeconomic policy trilemma, also called the impossible trinity. It explains the components of the impossible trinity: free capital mobility, exchange-rate management and an independent monetary policy and examines the impossibility of achieving all three components simultaneously. The unit also digs deeper into impossible trinity by analysing the empirical evidence. It also points out the problem of rupee convertibility in meeting the conditions of the impossible trinity.

Unit 10: **Approaches to Balance of Payments (BoP)** throws light on various approaches to BOP adjustment: The Elasticity Approach, the Absorption Approach, the Keynesian approach; and the monetary approach. The unit delves deeper by explaining these approaches and highlighting their importance in making BOP policy.

UNIT 7 EXCHANGE RATE REGIMES *

Structure

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7.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain the concepts of:
foreign exchange,
exchange rate, and exchange rate regimes;
- distinguish between different forms of exchange rate regimes;
- discuss the advantages and disadvantages of: fixed, floating, and managed float;
- trace briefly the historical development of international exchange rate regimes;
- examine operational aspects of the current exchange rate regimes; and
- provide an overview of India's exchange rate regime.

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7.1 INTRODUCTION

The exchange rate regime greatly impacts world trade and financial flows. The volume of such transactions and the speed at which they are growing highlight the crucial role of the exchange rate in today's world, thereby making the exchange rate regime a central piece of any national economic policy framework. The choice of an appropriate exchange rate regime—floating, managed or fixed arrangements—for individual countries is important. After all, an exchange rate regime that looks soft to one observer may look hard to another—which reflects, among other things, a lack of information among different players about foreign exchange markets and purchases or sales of foreign exchange by central banks. Which exchange rate regime and associated policies are appropriate for a country depending on its circumstances? No country can afford to be entirely indifferent to the behaviour of its exchange rate as it is closely related to the country's monetary policy, with both depending on common factors of influence and impact. In this unit, we discuss various aspects of the foreign exchange regime.

7.2 CONCEPTS

It is important to understand terms such as 'foreign exchange', 'exchange rate', and 'exchange rate regime' as they are central to understanding the economy around.

Foreign Exchange: 'Foreign exchange' refers to money denominated in a currency other than the domestic currency. Foreign exchange can be cash, funds available on credit cards and debit cards, and bank deposits.

Exchange rate: The rate at which a currency of one country exchanges for another country's currency is called the '*exchange rate*'. The exchange rate can either be expressed in terms of the number of units of domestic currency per unit of foreign currency (direct quotation) as in the case of most currencies such as the Indian rupee. For example, if the exchange rate between the rupee and the US dollar (USD) is quoted as Rs.65, this means that Rs.65 is required to purchase US\$1.00. Or the number of units of foreign currency per unit of domestic currency (indirect quotation) as in the case of some major trading currencies such as the pound sterling and the Australian dollar. When the value of the domestic currency increases in terms of another currency, it is referred to as a nominal appreciation of the domestic currency. In contrast, a decrease in the value of the domestic currency in terms of a foreign currency is known as nominal depreciation.

Exchange rate regime: An exchange rate regime is the process by which a country manages its currency with respect to foreign currencies. Exchange rate regimes can broadly be categorized into two extremes: fixed and floating. Between these, there are several combinations of the two. The exchange rate system refers to the arrangement for the movement of the exchange rate. Countries in the world operate under different exchange rate regimes.

7.3 IMPORTANCE OF FOREIGN EXCHANGE FOR THE ECONOMY

The exchange rate is a key financial variable that affects decisions made by foreign exchange investors, exporters, importers, bankers, businesses, financial institutions, policymakers and tourists in the developed and developing world. Exchange rate fluctuations affect the value of international investment portfolios, the competitiveness of exports and imports, the value of international reserves, the currency value of debt payments, and the cost to tourists in terms of the value of their currency. Thus, exchange rate movements have important implications for the economy's business cycle, trade and capital flows and are therefore crucial for understanding financial developments and changes in economic policy.

The exchange rate performs the same function as other prices in a mixed economy market. It provides information and incentives to guide decisions about what to produce and what to consume. Considered as a price, however, the exchange rate has special qualities. It is probably the single most important price in the economy. It affects numerous other prices and touches the interests of many people. This makes it the centre of much controversy with major exchange rate changes inviting popular attention. The exchange rate further links the general level of prices in the national economy with prices in other countries.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Define Exchange Rate.

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2) Why do exchange rates exist?

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3) Why is it important to study the exchange rate regime?

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7.4 EVOLUTION OF INTERNATIONAL EXCHANGE RATE REGIMES

The first modern international exchange rate regime was the gold standard. The adoption of the Gold Standard was a gradual process. There was no international legal foundation—no treaties, agreements or conferences. The various versions of the gold standard used were:

- a) gold specie standard,
- b) gold bullion standard, and
- c) gold exchange standard.

Fiat currencies are the norm in today's national economies and the current exchange rate regime. Stages in the development of an exchange rate regime may be grouped under the following *four* heads:

7.4.1 The pre-World War I (WWI) financial order (1870–1914)

The *gold standard* widely adopted in this era rested on converting paper notes into pre-set quantities of gold. From the 1870s to the outbreak of World War I in 1914, the world benefited from a well-integrated financial order, sometimes known as the *first age of globalisation*. Transactions were facilitated by widespread participation in the gold standard by independent nations and their colonies. This era saw mostly steady growth and a relatively low level of financial crises. The pre-World War I financial order was not created at a single high-level conference; rather it evolved organically in a series of discrete steps.

7.4.2 Between the World Wars (1919–1939)

The years between the world wars have been described as a period of *de-globalisation*, as both international trade and capital flows shrank compared to the period before World War I. During World War I, countries had abandoned the gold standard and, except for the United States, returned to it only briefly. By the early '30s, the prevailing order was essentially a fragmented system of floating exchange rates. In this era, the experience of Great Britain and others was that the gold standard ran counter to the need to retain domestic policy autonomy.

By the end of World War I, Great Britain was heavily indebted to the United States, allowing the USA to largely displace her as the world's number one financial power. As the *Great Depression* intensified in 1930, financial institutions were hit hard along with trade. Frequent changes in exchange rates introduced considerable uncertainty and made things difficult for the smooth conduct of international trade. This was amply proved by the experience of countries during the inter-war period, i.e., 1919-1939, which was marked by:

- i) competitive devaluation of currencies;
- ii) chaos in foreign exchange markets;
- iii) the decline in world trade;

- iv) imposition of trade restrictions; and
- v) widespread unemployment.

The lessons of the inter-war experience with exchange rates are:

- a) exchange rates should be reasonably stable for the smooth conduct of international trade;
- b) this can be achieved if countries, particularly advanced countries, cooperate; and
- c) International cooperation is the basis of a sound international monetary system or international exchange arrangement.

7.4.3 The Bretton Woods Era (1945–1971)

The principal architects of the new system, John Maynard Keynes and Harry Dexter White created a plan which was endorsed by the 42 countries attending the 1944 Bretton Woods conference, formally known as the United Nations Monetary and Financial Conference. The plan involved nations agreeing to a fixed but adjustable exchange rate system where the currencies were pegged against the dollar, with the dollar itself convertible into gold. So in effect, this was a gold-dollar exchange standard. There were several improvements to the old gold standard. Two international institutions, the International Monetary Fund (IMF) and the World Bank were created. The new exchange rate system allowed countries facing economic hardship to devalue their currencies by up to 10% against the dollar (more if approved by the IMF) – thus they would not be forced to undergo deflation to stay in the gold standard. A system of capital controls was introduced to protect countries from the damaging effects of capital flight and to allow countries to pursue independent macroeconomic policies while still welcoming flows intended for productive investment. However, pressing circumstances caused President Nixon to end all convertibility into gold on 15 August 1971. This event marked the effective end of the Bretton Woods system resulting in a system of floating exchange rates.

7.4.4 The post-Bretton Woods system (1971–present)

President Nixon is credited with ending the Bretton Woods Accord, and fixed rates of exchange, eventually creating a free-floating currency system. The current era has seen huge and turbulent flows of capital between nations. The transition away from Bretton Woods was marked by a switch from a state led to a market-led system. The Bretton Woods system is considered by economic historians to have broken down in the 1970s. Crucial events being:

- i) Nixon suspended the dollar's convertibility into gold in 1971,
- ii) the United States abandonment of capital controls in 1974, and
- iii) the UK's ending of capital controls in 1979 which was swiftly copied by most other major economies.

The contemporary exchange rate regime popularly known as ‘**non-system**’ is profoundly different from those envisaged at the 1944 meeting at Bretton Woods. The IMF officially endorsed the non-system at a meeting in Jamaica in January 1976. In the current system, exchange rates among the major currencies (principally the U.S. dollar, the euro, and the Japanese yen) fluctuate in response to market forces, with short-run volatility and occasional large medium-run swings. The IMF's approach continues to advise member countries on the implications of adopting different exchange rate regimes, consider the regime's choice to be a matter for each country to decide, and provide policy advice that is consistent with the maintenance of the chosen regime.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Briefly outline the historical development of the exchange rate regime before World War II.

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- 2) Examine the exchange rate regime from the Bretton Woods era to the present.

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7.5 FORMS OF EXCHANGE RATE REGIME

7.5.1 Fixed exchange rate

Under a fixed exchange rate system, a country's monetary authority agrees to manage its affairs to maintain a fixed ratio between the value of its own currency and that of other countries. A country's currency, therefore, has a definite objective value in terms of the other currencies into which it can be freely converted. The domestic currency is tied to another foreign currency, mostly more widespread currencies such as the U.S. dollar, the euro, the Pound Sterling or a basket of currencies. The government (or the central bank acting on the government's behalf) intervenes in the foreign exchange market to ensure that the exchange rate stays close to a predetermined target.

Under this system, exchange rate stability is achieved but if the exchange rate is fixed at the wrong rate, it may be at the expense of domestic economic stability. In a fixed exchange rate system, a rise in the exchange rate of the domestic currency *vis-à-vis* another foreign currency is called a *devaluation*. This means

that more of the domestic currency is needed to buy 1 unit of a given foreign currency. On the other hand, when the exchange rate falls, it is termed as a *revaluation*. These terms imply a deliberate decision on the government's part to change the exchange rate level. For example, a government's policy decision to devalue the domestic currency *vis-à-vis* the US dollar from Rs.65 to Rs.70.

It is important to realize that fixity cannot be achieved by simple government decree. There are powerful economic forces at work in the foreign exchange markets; restricting the movement of currencies requires equally decisive countervailing action by the authorities. This can take two broad forms: direct intervention in the foreign exchange using so-called *open market operations*, or the less direct option of *interest rate manipulation*. Fixing an exchange rate very rarely involves establishing a single point away from which a currency is not permitted to move. The usual approach is to define a *target zone* for the currency. The authorities then respond with appropriate measures when market forces threaten to move the currency above or below the zone. Theoretically, then, it is possible to hold a currency at any given rate, so long as the associated excess demand or supply of the currency at that rate can be met. The other policy measure that can be used as an alternative or supplement to open market operations involves the noted use of interest rates.

Advantages

- i) Provides greater certainty for exporters and importers as there are no or limited exchange rate risks. Hence it promotes long-term capital flows.
- ii) There is no fear of the adverse effect of speculation on the exchange rate.
- iii) Businesses have the knowledge that the price is fixed and therefore not going to change; it is relatively easier for them to plan.
- iv) Encourages international trade by making prices of goods involved in trade more predictable. It leads to low inflation.
- v) Imposes the discipline necessary for exchange rate stability; monetary policy must be coordinated. Keynes was a chief architect of Bretton Woods and a strong proponent of fixed rates. Many Keynesians agree that government intervention is necessary to promote exchange rate stability. However, the question remains how to attain a stable exchange rate system?

Disadvantages

- i) Have a high administration cost. Needs a complicated exchange control mechanism which may lead to the misallocation of resources.
- ii) A significant gap between the official rate and that determined by the market can promote black markets. In a black market, most foreign exchange transactions are carried out outside the banking system.
- iii) This may force the government to draw down on reserves to meet its obligations and cause a scarcity of foreign exchange.

- iv) Restricts the ability of countries to conduct policy autonomously.
- v) Inimical to the very desirable objective of free trade.
- vi) May achieve exchange rate stability but at the expense of domestic economic stability.

7.5.2 Floating exchange rate

A flexible system follows free market principles. Exchange rate determination is left entirely to the currency supply and demand processes, and governments do not attempt to manipulate the market to have particular exchange rate outcomes. *Popular regimes run the gamut from currency boards and traditional pegs to crawling pegs, target zones and floats with varying degrees of intervention.* It is closely related to monetary policy and the two are generally dependent on many of the same factors. The broad system of flexible exchange rates prevailing in the world economy since 1973.

Since market forces determine the exchange rate, the upward and downward movements in the value of the rupee are *appreciation and depreciation*. *Depreciation of the rupee* refers to the decrease in the external value of the domestic currency that occurred due to the operation of market forces. *Appreciation of the rupee* refers to the increase in the external value of the domestic currency that occurred due to the operation of market forces. The exchange rate is determined in the open market through the pressure of buying and selling foreign currencies.

Advantages

- i) A flexible exchange rate system confers several advantages upon economies that adopt it:
- ii) Continuous changes are easier to adjust to.
- iii) Less likely to permit a payments crisis to develop. Appropriate corrective exchange rate movements will swiftly dissipate any early or 'incipient' deficit or surplus that might arise. This system *automatically* provides for a balance of payments stability without the need for any action whatsoever by policy-makers.
- iv) Less politicized and authorities do not intervene in the market for foreign exchange and there is minimal need for international reserves.
- v) Less likely to have adverse domestic repercussions. Provides fewer incentives for destabilizing speculation. Financial instruments are available to hedge the risks posed by a fluctuating exchange rate.
- vi) Allows nations to pursue their own *independent* economic goals in respect of the other objectives of macro policy. Gives the government/monetary authorities flexibility in determining interest rates. This is because interest

rates do not have to be set to keep the value of the exchange rate within pre-determined bands.

- vii) Offers the most appropriate framework for the international allocation of resources through the trade process.

Disadvantages

- i) Market forces may fail to determine the appropriate exchange rate; hence, the floating exchange rate regime may not provide the desired results and may lead to the misallocation of resources.
- ii) It is impossible to have an exchange rate system without official intervention. The government may not intervene, however domestic monetary policy and fiscal policy would influence the exchange rate.
- iii) A wildly fluctuating exchange rate is at the mercy of national and international currency speculators. A volatile exchange rate introduces considerable uncertainty in export and import prices and consequently to economic development. At the same time, the abolition of exchange controls causes capital flight.

7.5.3 Managed floating

Many countries are practising a system of the managed floating exchange rate at present. The central banks have been trying to control fluctuations of exchange rates around some 'narrow band', however, the demand and supply forces determine the exchange rate. The decision to vary exchange rates can occur in the context of a variety of international monetary payment schemes. Countries may operate at a fixed rate but allow occasional variations for fundamental disequilibria. This is the *adjustable peg system* whereby rates can be pegged at new levels. Countries may allow rates to vary freely within agreed bands. Countries may allow unrestricted variation of rates. This allows a country to dispense with holding foreign reserves which represent a sacrifice of consumption.

In this hybrid exchange rate system, the exchange rate is determined in the foreign exchange market through the operation of market forces. Market forces mean the selling and buying activities by various individuals and institutions. The management of a floating rate is sometimes referred to in the literature as a '*dirty float*'. It is unlikely that the authorities in any country could remain completely indifferent to the behaviour of the exchange rate, and yet this is what the advocate of completely flexible rates recommends. Managed or dirty floating is the more usual alternative to participation in a fixed exchange rate system. In practice, a managed fiat permits the authorities to intervene in the foreign exchange markets, using direct open market operations, interest rates, or some combination of both.

In this context, the following table is very instructive:

Table 7.1: Forms of exchange rate regime

Fixed (pegged) Floating Exchange Exchange Rate Rate	Adjustable Peg	Crawling Peg	Managed Float	Wider Freely Band System		
↓	↓	↑	↑	↑	↑	↓
A						B

In the Table, as one moves from point A on the left to point B on the right, both the frequency of intervention by domestic monetary authorities and the required level of international reserves tend to be lower. Under a pure fixed-exchange-rate regime (point A), authorities intervene so that the value of the domestic currency vis-a-vis another country's currency says the US Dollar, is maintained at a constant rate. Under a freely floating exchange-rate regime, authorities do not intervene in the market for foreign exchange and there is minimal need for international reserves.

However, it remains true that there is no single exchange rate regime that is best for all countries in all circumstances. Countries continue to have the scope to choose the type of exchange rate regime that best suits their needs, always with the proviso that the chosen regime must be credibly supported by policies consistent with the choice. In light of important changes which include the general increase in capital mobility and the abrupt reversals of capital flows to developing and transition economies.

7.5.4 Recurrence of instability suggests a new regime

It is important to note that currently there have been shifting trends in the exchange rate regime. *After the breakdown of Bretton Woods, countries have adopted a wide variety of regimes, ranging from 'dollarisation' and currency boards to simple pegs and basket pegs, crawling pegs and target zones to clean floats and dirty floats.* There is such an abundance of possible linkages between the exchange rate regime and macroeconomic performance—some offsetting, others reinforcing—that, at a theoretical level, it is difficult to establish any unambiguous relationships at all. While increased capital mobility has been leading an increasing number of countries to either end of the spectrum between firmly fixed rates (or monetary unification) and free-floating, intermediate regimes are likely to remain viable and appropriate in many cases.

The recurrence of concerns and phases of instability suggests a need to look for durable remedies. Leaders continue to campaign for Bretton Woods II. It has been a feature of the **non-system** that exchange rates have at times, moved to positions that appear to be at variance with conditions prevailing in, and the longer-term interests of, domestic economies. Currently, the world economy is *generally* organised along flexible lines, but many countries choose to manage their currencies either alongside or in partnership with other currencies and countries.

There is a demand for *reform in the International Monetary System*. Zhou Xiaochuan, the former governor of the People's Bank of China proposed a gradual move towards increased use of **IMF special drawing rights (SDRs)** as a centrally managed global reserve currency. His proposal attracted much international attention. In December 2011, the Bank of England published a paper arguing for reform, saying that the current regime has performed poorly compared to the *Bretton Woods system*.

The steps required to move to an SDR-based system would be groundbreaking. A somewhat less ambitious alternative would be to make the SDR one of the alternative reserve currencies in a multicurrency system. Despite its relative stability, the current '*non-system*' has the inherent weaknesses of a setup with a dominant country-issued reserve currency, wherein the reserve issuer runs fiscal and external deficits to meet growing world demand for reserve assets and where there is no ready mechanism forcing surplus or reserve-issuing countries to adjust.

The current system has flaws, including occasional bouts of serious instability, but it has proved its strength and resilience when the conceivable alternatives have not. Charting the road ahead, the main task should be to help policymakers think beyond the status quo in fundamental ways. In this spirit, serious consideration should be given to alternatives to self-insurance, and to the potential—practical and political—for other currencies to acquire a greater role in the global reserve system. Such work could help the system evolve to shape developments rather than the other way around, as has historically tended to be the norm.

Neither the theory nor country experiences are very conclusive nor do exchange rate regimes have their benefits and costs. The suitability of a particular regime, however, depends on the policy target of a country.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) What are the main types of exchange rate systems? Does the choice of exchange rate regime matter?

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- 2) Explain the differences between fixed and flexible exchange rate systems and the advantages and disadvantages of each.

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- 3) What is Managed Floating Exchange Rate System?

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- 4) What are the main factors which determine the effectiveness of a floating exchange rate policy? Currently, which type of system determines the values of the major currencies, such as the Dollar, Yen, and Euro?

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- 5) What is a currency appreciation or depreciation? Point out the advantages and disadvantages of each.

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- 6) Explain how the choice of an exchange rate regime alters the conduct of monetary policy?

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- 7) Do you think the operation of non-system and recurrence of crisis needs a new regime for currency?

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7.6 INDIA'S EXCHANGE RATE REGIME

The main objectives of India's exchange rate policy are to ensure that the economic fundamentals are truly reflected in the external value of the rupee. In the post-independence period, India's exchange rate policy has shifted from a par value system to a basket-peg and to a managed float exchange rate system. The policy has evolved in line with the gradual opening up of the economy as part of the broader strategy of macroeconomic reforms and liberalization since the early 1990s.

India's rupee has matured to a regime of the floating exchange rate from the earlier versions of a '*managed float*'. The exchange rate policy is guided by the need to reduce excess volatility, prevent the emergence of speculative activities, help maintain adequate reserves, and develop an orderly foreign exchange market. The current exchange rate regime has been accompanied by full '*Convertibility on current account*' with effect from August 20, 1994. Moving forward, as India progresses towards full capital account convertibility and gets more and more integrated with the rest of the world, managing periods of volatility is bound to pose greater *challenges given the impossible trinity of independent monetary policy, open capital account and exchange rate management*. The principal features of the current exchange rate regime in India can be stated as follows:

- i) The rates of exchange are determined in the market.
- ii) The freely floating exchange rate regime continues to operate within the exchange control framework.
- iii) RBI can intervene in the market to modulate the volatility and sharp depreciation of the rupee.
- iv) The US dollar is the principal currency for RBI transactions.
- v) The RBI also announces a Reference Rate based on the quotations of select banks in Bombay at noon every day. The Reference Rate applies to SDR transactions and transactions routed through the Asia Clearing Union.

Check Your Progress 4

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Trace the historical development of the exchange rate regime in India.

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- 2) Discuss some of the features of the current exchange rate system in India.

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7.7 LET US SUM UP

In this unit we discussed about various exchange rate regimes. Exchange rate regime is an integral part of every country's economic and monetary policy. An exchange-rate regime is a way an authority manages its currency in relation to other currencies and the foreign exchange market. There are three types of exchange rate systems globally: flexible or floating exchange rate systems, fixed exchange rate systems and managed floating (intermediate exchange rate

system). Under **fixed exchange rates**, governments decide to fix their rates to some other standard, either a dominant currency such as the dollar, or a basket of currencies or gold. Under floating exchange rates, the market determines the outcomes of rates. In a **floating exchange rate regime** the market's demand for and supply of the currency determines the exchange rate. There is no pre-determined official target for the exchange rate set by the government. The latter and/or the central bank can indirectly influence the exchange rate by managing the level of domestic and foreign currencies in the banking system. In managed exchange rate regimes, the value of the currency is determined to vary degrees by government intervention. A country can alter the degree to which it intervenes in the foreign exchange market. **Managed float** is also referred to as **dirty float**. The exchange rate is the dominant target of economic policymaking.

Balance of payments on current account disequilibrium can automatically be restored to equilibrium through a floating exchange rate regime and the scarcity or surplus of any currency is eliminated. Balance of payment adjustment is smoother and painless under a floating exchange rate regime compared to a fixed exchange rate system. The autonomy of monetary authorities is preserved under a floating exchange rate system as there is no target exchange rate to maintain. Since 1971 exchange rate system is known as a **non-system**. This variety of exchange rate regimes exists in today's global financial environment.

The history of exchange rate regimes can be categorised into four parts in the following way:

- i) Before World War I (WWI) financial order (1870–1914)
- ii) Inter-war period (1919–1939)
- iii) The Bretton Woods system (1945–1971).
- iv) The post-Bretton Woods system (1971–present): the non-system.

In the post-independence period, India's exchange rate policy has seen a shift from a par value system to a basket-peg and further to a managed float exchange rate system during the period 1947 to 1971. In March 1993 the system of the market-determined exchange rate was adopted. However, the RBI did not relinquish its right to intervene in the market to enable orderly control.

7.8 KEY WORDS

The black market for foreign exchange

is a market where foreign currencies are exchanged for the national currency of a country at a premium over the rate at which the value of the national currency is pegged by its monetary authority. A black market for foreign exchange is an outcome of an over-valued pegged exchange rate

regime together with binding exchange control.

BoP crisis

is a situation where foreign currency reserves held by a country's central bank dry up and the country defaults on payment for its imports and debt repayment. A BoP crisis emerged for India in early 1991 when it had foreign currency reserves that could barely finance a week's imports.

Clean float

a regime where a country's monetary authority does not intervene in the foreign exchanges market in any way whatsoever and allows the price of its own currency in terms of a particular foreign currency to adjust to market conditions.

Conventional Pegs

are those in which the authorities intervene to keep the exchange rate at a pre-arranged level.

Convertibility of rupee on capital account

means the freedom to convert rupees into foreign currency and back for capital transactions. It is a feature of a nation's financial regime that centres on the ability to conduct transactions of local financial assets into foreign financial assets freely or at country-determined exchange rates. It is sometimes referred to as capital asset liberation or CAC.

Convertibility of rupee on current account

implies that the Indian rupee can be converted to any foreign currency at existing market rates for trade purposes for any amount. Anyone who deals in the current account means international trade of goods and services can convert them to Indian Rupees.

Crawling Pegs

when the exchange rate is regularly adjusted in the same direction by a pre-arranged amount.

Currency Boards

the regime under which the *only* role of monetary policy is to keep the domestic currency's value constant relative to a particular foreign currency.

Devaluation

is a deliberate downward adjustment to the value of a country's currency relative to another currency, group of currencies or standard. Devaluation is a monetary policy tool used by countries that have a fixed exchange rate or semi-fixed exchange rate. It is often confused with depreciation and is the opposite of revaluation.

Dirty float

is a currency that floats in value in terms of other currencies but is not free of government intervention. Governments intervene to smooth fluctuations or manage a desired exchange rate.

Dollarization

involves one country unilaterally adopting the currency of another.

Exchange control

is a cap imposed by the monetary authority of a country on foreign currency buying by its domestic importers and other buyers.

Full Convertibility of rupee

fully convertible, when there are no current account or capital account restrictions for free movement of funds into and out of the country. As there are a lot of restrictions for the free movement of funds from India and also into the country, Indian Rupee is not fully convertible.

Gold standard

was an international monetary system in which gold was the reserve asset against coins and currencies in circulation in different countries. With the gold standard, countries agreed to convert paper money into a fixed amount of gold. A country that uses the gold standard sets a fixed price for gold and buys and sells gold at that price.

International currency system

defines the parameters of the foreign exchange market where national currencies are exchanged and traded.

Managed Floating Rates

these are the rates under which authorities intervene to limit short-term currency fluctuations without the goal to keep the exchange rate at a certain level.

Special Drawing Rights (SDRs)	IMF created an international reserve asset in 1969 to support the Bretton Woods fixed exchange rate system. It is defined as the weighted average of four major currencies – euro, Japanese yen, pound sterling, and USD.
The Bretton Woods system	was a gold-USD exchange standard that defined the international currency system during 1944-71. Currencies of different countries were pegged to the USD, which, in turn, was convertible into gold at USD 35 per ounce.

7.9 SOME USEFUL REFERENCES

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7.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check your progress 1

- 1) see section 7.2
- 2) see section 7.2
- 3) see section 7.3

Check your progress 2

- 1) see sections 7.4.1 and 7.4.2
- 2) see sections 7.4.3 and 7.4.4

Check your progress 3

- 1) See section 7.5.1 and 7.5.2
- 2) See section 7.5.1 and 7.5.2
- 3) See section 7.5.3
- 4) See section 7.5.3 and 7.5.4
- 5) See section 7.5.2
- 6) See section 7.5.3 and 7.5.4
- 7) See section 7.5.3 and 7.5.4

Check your progress 4

- 1) See section 7.6
- 2) See section 7.6

UNIT 8 COMPONENTS OF BALANCE OF PAYMENTS *

Structure

- 8.0 Objectives
- 8.1 Introduction
- 8.2 Importance of balance of payments (BoP) for a country
- 8.3 Concept of BoP
- 8.4 Some related concepts
 - 8.4.1 Balance of trade and Balance of payments
 - 8.4.2 Autonomous and accommodating transactions
 - 8.4.3 Balance of payments always balances
- 8.5 Components of BoP
- 8.6 BoP Accounting: An example of India's BoP
- 8.7 Nature and implications of disequilibrium
- 8.8 Policy measures for correcting disequilibrium
- 8.9 Let us sum up
- 8.10 Key Words
- 8.11 Some Useful Books
- 8.12 Answer/Hints to Check Your Progress Exercises

8.0 OBJECTIVES

After studying this Unit, you should be able to:

- appreciate the importance of the balance of payments (BoP) for a country;
- explain the concepts related to BoP;
- distinguish between:
 - i) balance of trade and balance of payments;
 - ii) current and capital account of BoP;
 - iii) autonomous and accommodating transactions of BoP;
- describe the components of the current and capital account of BoP;
- highlight the significance of autonomous and accommodating items;
- discuss the nature and implications of disequilibrium in the form of 'surplus' and 'deficit' of BoP; and
- examine the policy measures required for removing disequilibrium in BoP.

* Contributed by Prof. S.K. Singh, Retd. Prof. IGNOU

8.1 INTRODUCTION

This unit examines the framework used to summarize a country's international transactions. The scorecard is the balance of payments, the set of accounts recording all flows of value between a nation's residents and the rest of the world during a period of time. The balance of payments shows a wealth of information about a country's international economic transactions. An international transaction is an exchange of goods, services, or assets between residents of one country and those of another. It is also a key to understanding how people trade one country's money for that of another country. In addition, the exchanges documented in the balance of payments have major implications for macroeconomic concerns like growth, inflation and unemployment. Trends in the balance of payments as a whole, including trade, need to be analyzed to assess the interactions and inter-relations between an economy's domestic and external sectors.

8.2 IMPORTANCE OF BALANCE OF PAYMENTS (BoP) FOR A COUNTRY

BoP is one way to measure a country's international economic situation. It indicates a country's status in international trade. Persistent disequilibrium in the balance of payments, particularly the deficit balance, is undesirable because it (a) weakens the country's economic position at the international level, and (b) affects the economy's progress adversely. BoP accounting highlights a country's competitive strengths and weaknesses and helps achieve balanced economic growth. It can significantly affect a government's economic policies and the economy itself. Capital receipts and payments indicate the country's external debt position and changes in it. BoP data are of interest because they reveal demand for the country's currency and BoP trend helps to predict what sort of economic environment may develop in the country.

The transition to floating exchange rate regimes in the 1970s changed the focus from the total BoP to its various subaccounts like the Current and Financial Account balances. These are the indicators of economic activities and currency repercussions to come. The balance of payments can be used to forecast economic conditions and, in some cases, the likelihood of an economic crisis. International crises have occurred for as long as there have been international trade and commerce. And they will occur again. Each crisis has its unique characteristics, but all follow some of the economic fundamentals.

8.3 CONCEPT OF BoP

The balance of payments of a country is a systematic record of all economic transactions during the period between residents of the reporting country and residents of other countries. It is a summary of the monetary value of all exchanges and transfers of goods, services and evidence of debt or ownership –

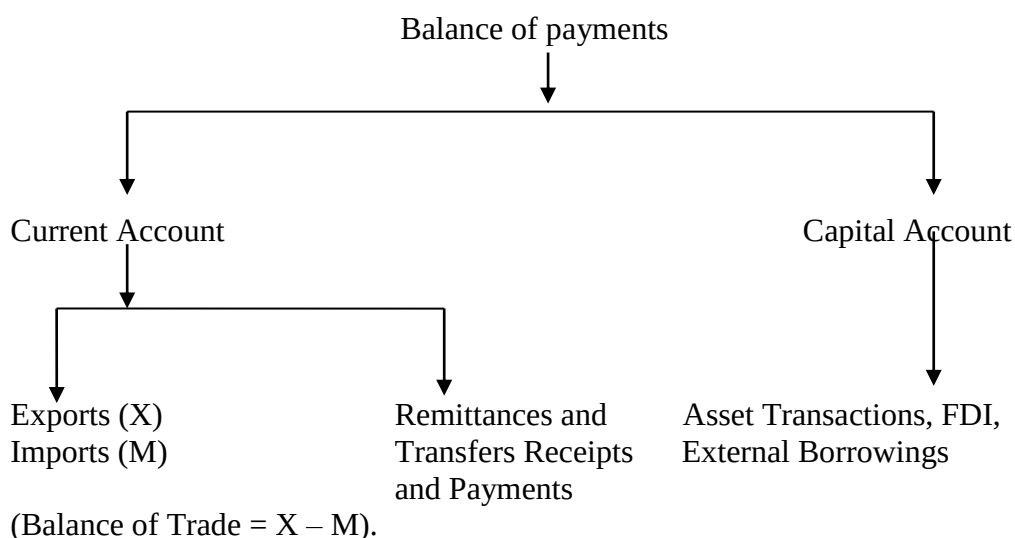
suitably classified and aggregated between the residents, businesses and government and other institutions of one country and the rest of the world for a given period of time, usually a year. Exports and imports of goods and services, and all the other items, are flow items. That has all dimensions of an amount per day, month, year, or whatever the period in question may be. The International Monetary Fund (IMF) has developed a standardized BoP system and form of presentation. All transactions entering the balance of payments can be grouped into three broad accounts within the balance of payments:

- i) the current account - records transactions in goods and services and transfer payments;
- ii) the capital account - represents a change in a country's foreign assets and liabilities; and
- iii) the official international - records transactions in monetary gold and other reserve assets (notably convertible currencies, the country's 'reserve tranche' at the IMF and SDRs.

The BoP is often misunderstood because many people believe it to be a balance sheet, rather than a cash flow statement. By recording all international transactions over a period of time, it is tracking the continuing flow of purchases and payments between a country and all other countries. It does not add up the value of all assets and liabilities of a country like a balance sheet does for an individual firm. In short, BoP means:

- i) record of all international economic transactions;
- ii) the type of economic transactions that dominate are real assets, financial assets, and services;
- iii) it is not a balance sheet;
- iv) it is a flow statement; and
- v) during a period of time, usually one year.

In a tabular form structure of BoP transactions may be presented as under:



8.4 SOME RELATED CONCEPTS

For the sake of clarity in understanding, three important concepts are explained in this section. They are: (i) the distinction between the balance of trade and the balance of payments, (ii) autonomous and accommodating transactions, and (iii) the balance of payments always balances.

8.4.1 Balance of Trade and Balance of Payments

A marked distinction exists between the concepts of balance of trade and balance of payments. BoP transactions take the form of trade in goods and services, capital movements or financial flows. These three forms of transaction give rise to a compartmentalization of the balance of payments accounts. Balance of trade refers to the visible items only. Import or export of goods is a visible item because it is an open trade among the countries and can be easily certified 'by the customs officials'. The net receipts from exports and imports of merchandise constitute the balance of trade (BoT). A trade deficit for a country arises in a particular year if the value of its exports is smaller than the value of its imports. The balance of trade need not always balance. The important point to remember is that there may be a balance of *trade* deficit and still a balance of *payment* surplus or *vice versa*. In other words, a trade deficit does not necessarily imply that the country is losing reserves, etc.; long-term movements account for the difference.

The merchandise trade is only a portion of the thousands of different international transactions that occur in the country each year. It is the official term for net exports that makes up the balance of payments. It is just one part of the balance of payments. The official balance of trade is separated into the balance of merchandise trade for tangible goods and the balance of services. BoP is more comprehensive than the balance of trade. Balance of payments includes apart from the balance of trade or merchandise account, the invisible account which is again composed of the services sector and the gifts and charities account comprising a variety of invisible items, plus a record of the capital account. Thus, the balance of trade is only a partial study of the balance of payments. However, the balance of trade is nothing but a major component of the balance of payments. In short, the balance of trade is a partial picture, while the balance of payments is a complete picture of the country's international economic relations. Further, in the accounting sense, the balance of trade may be a deficit or surplus. It may, thus, be imbalanced. But the balance of payments as a whole must always balance. For that reason, there is an item like "Errors and Omissions" in its structure.

8.4.2 Autonomous and accommodating transactions

A double-entry accounting principle leads to the distinction generally made between underlying, originating or '*autonomous*' transactions, and those which are merely the financial counterparts or the '*accommodating*' transaction. This distinction is usually attributed to *James Edward Meade*. Autonomous transactions such as trade flows, direct investment, and portfolio investment are independent of other items and not restricted by the size of other balance of payments transactions. These items are also called *above-the-line items in BoP accounts*. Autonomous transactions are undertaken solely to make international trade possible. '*Autonomous*' refers to transactions undertaken spontaneously for their own sake.

Accommodating transactions depend on the other items and should be shown as identically equal to the autonomous transaction such that total credits equal total debits. Accommodating items reflect government actions aimed at altering the size and the composition of the balance of payments, and they are sometimes referred to as *below-the-line items*. Representing each entry independently will not guarantee equality between credits and debits. As such, it is important to identify the counterpart entries or '*accommodating*' items to ensure total credits equal total debits. It should be clear that net autonomous transactions are counterbalanced by equivalent net accommodating transactions. A balance of payments surplus or deficit refers only to autonomous transactions.

8.4.3 Balance of payments always balances

The *balance of payments* is strictly an accounting term used to describe *a record of all of the transactions between one country and the rest of the world*. A *balance of payments problem* is, accordingly, *a problem about how the balance of payments balances*. In practice, it is difficult to achieve where receipts equal payments. In reality, total receipts may diverge from total payments because of:

- i) the difficulty of collecting accurate trade information;
- ii) the difference in the timing between the two sides of the balance; and
- iii) a change in the exchange rates.

To say that the '*balance of payments always balances*' is to say that a net credit balance in one of these accounts must have a counterpart net debit balance in one of the other accounts. The algebraic sum of the three accounts' net credit and debit balances must equal zero. Thus, a net credit balance on the current account of say, (+) Rs. 100 crores, may have a counterpart net debit balance on the capital account of say (-) Rs.100 crores (that is, an outflow of capital), a net inflow of gold by Rs.100 crores or a combination of capital outflow and gold inflow equal to the Rs. 100 crores surplus on the current account. It simply results from the double-entry book-keeping procedure which is used to record the transactions.

Because of measurement problems, recourse is made to '*balancing item*' that intends to eliminate measurement errors. The purpose of incorporating this item in the BoP account is to adjust the difference between the sums of the credit and the sums of the debit items in the BoP accounts so that they add up to zero by construction. Hence the proposition: '*the BoP always balances*'. It is a truism. It only suggests that the two sides of the accounts must always show the same total. It implies only equality. In this bookkeeping sense, BoP always balances. '*Balance of payments always balances*' has no economic significance.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Define the balance of payments.

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2) Distinguish between autonomous and accommodating transactions in the balance of payments.

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3) Why must a country's balance of payments always be balanced in theory?

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4) How does the balance of payments differ from the balance of trade?

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5) Highlight the importance of BoP for an economy.

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8.5 COMPONENTS OF BoP

There are a wide variety of accounts that determine the BoP, but for purposes of analysis, they can be grouped into three broad categories: current account items, capital account items, and reserves.

Current account

The current account consists of *merchandise trade, services, and unrequited transfers*. Merchandise trade is typically the first part of the current account. It receives more attention than any of the other accounts because this is where *the imports and exports of goods are reported, and these are often the largest single component of all international transactions. Merchandise trade transactions can affect a country's BoP in several ways.* The transactions in the current account affect national income and, in terms of income analysis, the level of output and employment.

The second type of transaction in the current account is a payment (or receipts) for which there are no corresponding receipts (or payments). These are called unrequited payments and receipts. These transactions are unilateral transfers. The net receipts from these transactions and the transactions in the trading account constitute the balance of the current account. Thus, positive net receipts imply a current account surplus and negative net receipts-payments exceeding receipts imply a current account deficit.

Services

The services category includes many payments such as freight, banking and insurance on international shipments; tourist travel; profits and income from overseas investment; personal expenditures by government, civilians, and military personnel overseas; and payments for management fees, royalties, film rental, and construction services. *Purchases of these services are recorded as debits, while sales of these services are similar to exports and are recorded as credits.*

The balance of the current account, is a larger concept than the balance of trade, as it includes the balance of trade, the balance of services, and the balance of unrequited transfers. The balance of the current account need not be equal but can show a surplus or a deficit. The balance of the current account is a very important concept, as it shows the flow aspect of a country's international transactions.

Capital account

The capital account records all international purchases and sales of assets such as money, stocks, bonds, etc. Capital account items are transactions that involve claims in ownership. It shows the net change in foreign-asset ownership of a nation. *Direct investment* involves managerial participation in a foreign enterprise and some control. *A portfolio investment* is an investment designed to obtain income or capital gains. Transactions in the capital account affect the country's international debtor or creditor position and the distribution of wealth and debt. Capital account transactions give rise to future claims such as acquiring foreign assets or shares in companies located abroad.

The capital account consists of the reserve account (the net change of foreign exchange of a nation's central bank in market operations), loans and investments made by the nation (excluding the future interest payments and dividends yielded by loans and investments). If net foreign exchange is negative, the capital account is said to be in deficit. The capital account transactions are difficult to measure. The capital and financial accounts could run a large enough surplus to offset a *trade deficit*.

Reserves or monetary movements

This refers to a record of the transaction with the International Monetary Fund (IMF) and foreign exchange reserve that mainly consists of holding gold and foreign currency assets. Drawings (treated as a kind of borrowing) from the IMF is a credit item, whereas repayments made to IMF are debit items. Drawing down of reserve, which is an inflow into the balance of payments from the reserve account, is a credit item. Like any other inflow, these reserves can be used to support a deficit elsewhere in the balance of payments. Similarly, additions to the reserve account are an outflow from the balance of payments to the reserve account and are, therefore, a debit item. Reserves are used for bringing BoP accounts into balance. When all the components of the balance of payments are taken together, the balance of payments should be in balance. Credits should equal debits.

Balancing item of BoP

Given that millions of rupees in transactions are reported in BoP statements, it should come as no surprise that the number of recorded debits is never equal to the number of credits. This is why there is an entry for net errors and omissions in the reserve account. Net errors and omissions reflect the imbalances resulting from imperfections in source data and compilation of the balance of payments accounts. This is also known as a balancing item.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Explain the components of BoP.

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2) What are merchandise trade and service trade?

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3) Distinguish between the current and capital account of BoP.

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4) What is the significance of balancing items of the BoP account?

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8.6 BoP ACCOUNTING: AN EXAMPLE OF INDIA'S BoP

Let us understand various components of BoP by having an example of India's BoP. The following table illustrates the items mentioned in BoP of India and a brief account of the items is explained below the table.

Table 8.1: India's Overall Balance of Payments - US Dollar

Item/Year	(US \$ Million)		
	2021-22		
	Credit	Debit	Net
1	17	18	19
A. Current account			
1. Merchandise	429164	618623	-189459
2. Invisibles (a+b+c)	369600	218906	150694
a) Services	254527	147011	107516
b) Transfers	89322	8875	80447
c) Income	25751	63020	-37269
Total Current account (1+2)	798764	837529	-38766
B. Capital account			
1. Foreign investment	478000	456191	21809
2. Loans	105457	71851	33605
3. Banking capital	102886	96217	6669
4. Rupee debt service	0	71	-71
5. Other capital	75059	51265	23794
Total capital account (1 to 5)	761402	675595	85807
C. Errors & omissions	1433	974	459
D. Overall balance	1561599	1514098	47501
E. Monetary movements (i+ii)	16024	63524	-47501
i) I.M.F.	0	0	0
ii) Foreign exchange reserves (Increase- / Decrease+)	16024	63524	-47501
G.n.i.e.: Government not included elsewhere.			
Notes: 1. Data for 2021-22 are preliminary estimates			
2. Total may not tally due to rounding off.			

Source: The Handbook of Statistics on the Indian Economy 2021-22, RBI, Mumbai

The Table 8.1, exemplifies India's BoP account for the financial year 2021-22. It shows the main items of transactions with some simplifications. The total receipts in the trading account show the balance from exports and imports of merchandise only. These include the physical movement of goods, i.e. manufacturing products, semi-finished goods, capital goods, raw materials, agricultural products, etc. During 2021-22, India has a deficit in trade account to the extent of -189459 million \$. This trade account deficit is because of excess merchandise imports (debit) over merchandise exports (credit). The next item is Invisibles which are divided into three components: (i) services, (ii) transfer payments, and (c) investment income.

Services include (a) travel and tourism; (b) transportation; (c) financial and other services including insurance; (d) government; (e) software services exports, etc.

Investment income refers to receipts (credit) and payments (debit) of dividends, interests, and profits arising out of Indian investments abroad and foreign investments in India.

Transfer payments do not represent payment for any direct service rendered or any physical transfer of goods. They are like foreign aid, gifts, foreign workers' remittances to their home countries, etc. Workers' remittances are sizable amounts (\$80447 million) and form a very important component of transfer payment accounts in India. It is to be noted that mainly on account of this a surplus of \$107516 million was achieved under this head during 2021-22.

The *current account* balance of \$ 38766 million (a sum of items I and II) refers to the balance in flows of goods (merchandise) and services and other current receipts and payments (investment income and transfer payments) between countries. India had a deficit in the trading account (item-I) and a surplus in the invisible account (item-II) but owing to the large deficit in the trading account India had a net deficit in the current account (item III) to the tune of \$ 38766 million.

The capital account is a miscellaneous account. The two main types of transactions in the capital account on the credit (or receipt) side are foreign investments and borrowing abroad. These are the import of capital, but unlike the import of goods and services, these are mentioned on the credit side because the import of capital means foreign exchange earned. The corresponding debit (or payments) items are investments abroad and leading. Taking all the items into account one gets total capital account payments and receipts. The table shows that India had a capital account surplus during 2021-22 to the tune of \$ 85807 million.

Despite transactions being recorded, based on the double-entry method discrepancies crop up between debits and credits because of data lags and other estimation problems. These discrepancies are captured under '*errors*' and '*omissions*'. A negative value indicates that receipts are overstated, payments are

understated, or both, and *vice versa*. The overall balance is obtained as a sum of the current account and capital account after adjusting for errors and omissions. India had an overall surplus balance of \$47501 million in 2021-22. An overall surplus means that India has a claim on the rest of the world.

The monetary movement account of BoP records changes in the amount of ‘official’ reserve assets. *Official* reserve assets include gold, foreign currencies, and the IMF’s special drawing rights (SDRs) reserve positions. The deficit means that the rest of the world has a claim on India, which is an asset for the rest of the world. The deficit is officially settled by corresponding entry on the credit side as import of capital or increase in foreign currency reserves.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Briefly explain the elements of BoP accounting.

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2) What do you mean by the monetary movement account of BoP?

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3) What is the overall balance?

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4) What are transfer payments?

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8.7 NATURE AND IMPLICATIONS OF DISEQUILIBRIUM

The problem of judging the equilibrium position of a country’s balance of payments becomes fairly complicated and takes outside the sphere of bookkeeping. According to *Czinkota*, the mechanics of disequilibrium in BoP often follow a similar path of development:

i) A country that experiences a rapidly expanding current account deficit will simultaneously build financial account surpluses.

- ii) The capital that flows into a company, giving rise to the financial account surplus, acts, as the '*financing*' for the growing merchandise/services deficits-the constituent components of the current account deficit.
- iii) Some events, whether it is a report, a speech, or an action by a government or business inside or outside the country, raises questions about the country's economic stability. Investors of many kinds, portfolio and direct investors in the country fearing economic problems in the near future, withdraw capital from the country rapidly to avoid any exposure to this risk. This is prudent of the individual but catastrophic for the whole if all individuals move similarly.
- iv) The rapid withdrawal of capital from the country, also known as '*capital flight*', results in the loss of the financial account surplus, creating severe deficits in the country's overall balance of payments. This is typically accompanied by rapid currency depreciation (if a floating-rate currency) or currency devaluation (if a fixed-rate currency).

International debt and economic crises have occurred for as long as there have been international trade and commerce. And they all will occur again. *Each crisis has its unique characteristics, but all follow some of the economic fundamentals.* There are several causes and each country will have a unique set of circumstances. In short, some of the common wider economic effects of the BoP crisis or disequilibrium are:

- i) need to import financial capital to achieve balance;
- ii) currency weakness and higher inflation;
- iii) may run short of vital foreign currency reserves;
- iv) lack of price/non-price competitiveness;
- v) can lead to capital flight/loss of investor confidence.

A current account deficit, if persistent, is not sustainable because, on the one hand, foreign capital may take a dim view of the country's ability to meet its foreign obligations and, therefore, cut down the flows, and on the other hand, monetary movements, particularly domestic reserve account may also find the deficit unmanageable and get drained. Since a deficit represents an imbalance between demand for and supply of foreign exchange, a persistent presence of this imbalance can destabilize the currency.

A persistent surplus in the current account is also not desirable because it means that either the country invests the surplus abroad for another country's development or it allows its currency to appreciate. The former does not add to GDP; the latter slows down GDP growth by crowding out exports. What is desirable is a current account surplus, as it enhances the capacity of the country to generate an excess of exports over imports, sufficient to pay for charges on account of interest or dividends on foreign capital.

8.8 POLICY MEASURES FOR CORRECTING DISEQUILIBRIUM

Several policies can be introduced to improve a country's trade balance. The measures are:

- i) Adjustment through Exchange Depreciation.
- ii) Devaluation or Expenditure-Switching Policy.
- iii) Direct Controls.
- iv) Adjustment through Capital Movements.
- v) Adjustment through Income Changes.
- vi) Stimulation of Exports and Import Substitutes.
- vii) Expenditure-Reducing Policies.

Some of these measures focus on changing the growth of demand while others look to improve an economy's supply-side competitiveness. As with any macroeconomic '*problem*,' effective policies are those that target the underlying causes. The real solution lies in earning more foreign exchange through additional exports or reducing imports. Quantitative changes in exports and imports require policy changes. Such policy measures are in the form of monetary, fiscal and non-monetary measures. Non-monetary methods are more effective than monetary methods and are normally applicable in correcting an adverse balance of payments.

There are two possible courses of action: *expenditure reduction policy* or *expenditure-switching policy* or some combination of both. The **expenditure reduction policy** aims to dampen the level of aggregate demand in the domestic economy so that the demand for imported goods and services is reduced. Unfortunately, dealing with a balance of payments deficit in this way has deleterious consequences for some of the other major macroeconomic objectives. Thus a lower level of demand is likely to constrain and affect an increase in unemployment. Balance of payments equilibrium should not be relentlessly pursued at the expense of other goals or objectives.

Expenditure-switching policies intend to switch domestic and foreign demand away from foreign goods and towards home-produced goods. The objective is to encourage overseas residents to buy more exports and to induce domestic residents to purchase home-produced goods and services instead of imports. This requires modifications in consumption behaviour. The most obvious method is to *devalue* the domestic currency. Currency devaluation induces relative price changes that favour the products of the devaluing economy. There must be a reasonable degree of price elasticity of demand for exports and imports. It is generally recognized that the price elasticity of demand for exports plus the price

elasticity of demand for imports must be greater than 1 (the noted Marshall-Lerner condition).

It should be noted that several additional measures are available to governments, each of which also has expenditure-switching properties. These include import controls, export promotion, and the promotion of domestic industry by fiscal or other means. A realistic exchange rate, low tariffs, and minimal trade transaction costs should guide exports. *To sum up*, the correction of disequilibrium calls for a judicious combination of the following measures:

- i) Monetary and fiscal changes affecting income and prices in the country;
- ii) Exchange rate adjustment, i.e., devaluation or appreciation of the home currency;
- iii) Trade restrictions, i.e., tariffs, quotas, etc.; and
- iv) Capital movement, i.e., borrowing or lending abroad.

No reliance can be placed on any single tool. There is room for more than one approach and more than one device. But the application of the tools depends on the nature of the disequilibrium. Countries can be in very different positions concerning the balance of payments. It is, important to understand a country's balance-of-payments situation intelligently. To understand the nature of a deficit one has to judge it against the background of the general economic policy of a country and the policy means a country has at its disposal. The country must in general try to implement policy measures aimed at reducing the deficit. Surpluses do not usually create great problems. International capital flows play an important part in equilibrating the balance of payments.

Check Your Progress 4

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) What are the common wider economic effects of BoP disequilibrium?

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- 2) Why is a persistent surplus in the current account not desirable? Give reasons.

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3) Why is a persistent deficit in the balance of payments undesirable?

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4) What policy measures are available for a country to achieve an improvement in a country's BoP?

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8.9 LET US SUM UP

In this unit, we discussed about various components of Balance of Payments. Balance of payments (BoP) is a record of the values of all economic transactions of a country with the rest of the world in a particular year. Two types of transactions dominate the balance of payments. One is *real assets* like the exchanges of goods and services and the other is *financial assets* consisting of the exchange of financial claims (for example, stock, bonds, loans, purchases or sales of companies) in exchange for other financial claims or money. *The Balance of Trade deals* only with the export and import of merchandise or visible items. It is just one part of the balance of payments.

The current account of the balance of payments refers to the monetary value of international flows associated with transactions in goods and services, income flows and unilateral transfers. Capital transactions consist of capital transfers and the acquisition and disposal of certain nonfinancial assets. Under *capital account*, there is only inflow and outflow of capital and the difference between the two represents a country's capital account balance.

Balance of payments accounts is constructed using a double entry accounting principle such that total credits equal total debits. Showing each credit and debit entry independently will not ensure that total credits equal total debits. Autonomous transactions are the transactions, undertaken for their own transactions. Accommodating transactions are the counterpart entries that total credits equal total debits and will usually be the financial counterpart or means of payment for the original autonomous transaction. Current account transactions are all autonomous, capital movements are capital account transactions can be both autonomous and accommodating transactions.

A deficit in BoP must be settled. A way to settle is by a transaction on the capital account. Contrary to the general perception, a current account deficit is not a sign of bad economic policy or bad economic conditions.

BoP is one of the major indicators of a country's status in international trade. International managers can use the balance of payments to forecast economic conditions and, in some cases, the likelihood of economic crises. The accounts do not tell what is good or bad, nor do they tell what is causing what. But they show what is happening so one can reach one's conclusions.

A deficit in BoP must be rectified by taking appropriate measures. There are many methods, important among them are monetary and non-monetary measures.

The non-monetary category includes trade policy for promoting exports and reducing imports. Capital inflows and Fiscal policy measures. Monetary measures include expenditure-switching policies which work through changes in relative prices. An important form of expenditure switching policy is the reduction in the foreign exchange rate of the national currency, namely, devaluation. The objective of expenditure-reducing policy in the economy relates to the fall in aggregate expenditure which works to reduce imports.

8.10 KEY WORDS

Accommodating transaction

One undertaken for the balance of payments purpose.

Autonomous transaction

One undertaken for its own sake.

Balance of payment (BoP)

This is a record of the value of all economic transactions of a country with the rest of the world for a particular year. It is the sum of current account and capital account transactions including the term of the official settlement.

Balance of trade

Refers to merchandise trade only i.e. receipts from exports less payment for imports of a country.

BoP account

Follows the double-entry bookkeeping method, which balances both the credit and debit sides. By this accounting method, the BoP of a country is always in balance.

Counterpart items

Certain items in the balance of payments exist only as counterpart items, introduced to balance the inclusion of other items that do not fall naturally into the double-entry system. The allocation of Special Drawing Rights (SDRs) is an example of an artificial counterpart item introduced into the balance of payments to offset the corresponding increase in SDRs holdings within official

reserves (as SDRs are no one sector's liabilities).

Expenditure reduction policy

Involves a reduction in aggregate demand in the domestic economy to improve the balance of payments position on the current account.

Expenditure-switching policy

Switches domestic and foreign demand away from foreign goods and towards home-produced goods.

Monetary movements

The items making monetary movements are as follows: miscellaneous capital, change in external liabilities and claims, change in the account with the I.M.F., and change in the gold and convertible currency reserves.

Rupee debt service

Interest payments on and principal repayments on account of civilian and non-civilian debt in respect of Rupee Payment Area (RPA), are clubbed together and shown separately under this item.

8.11 SOME USEFUL REFERENCES

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Roy, Shyamal (2010): *Macroeconomic Policy Environment*, 2nd edition, Tata McGraw-Hill Publication Company Ltd., New Delhi

Sodersten, Bo (1974): “*International Economics*”, Macmillan Press Ltd., London, UK.

8.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check your progress 1

- 1) See section 8.3
- 2) See section 8.4.2
- 3) See section 8.4.3
- 4) See section 8.4.1
- 5) See section 8.2

Check your progress 2

- 1) See section 8.5
- 2) See section 8.5
- 3) See section 8.5
- 4) See section 8.5

Check your progress 3

- 1) See section 8.6
- 2) See section 8.6
- 3) See section 8.6
- 4) See section 8.6

Check your progress 4

- 1) See section 8.7
- 2) See section 8.7
- 3) See section 8.8
- 4) See section 8.8

UNIT 9 IMPOSSIBLE TRINITY: ALTERNATIVE SCENARIOS *

Structure

- 9.0 Objectives
- 9.1 Introduction
- 9.2 The Concept of Impossible trinity
- 9.3 Theoretical underpinning: Mundell-Fleming model
- 9.4 Impossible Trinity: alternative scenarios countries' experience
- 9.5 Importance of Impossible Trinity
- 9.6 Impossible trinity and demand for capital account convertibility of India's rupee
- 9.7 Let us sum up
- 9.8 Key terms
- 9.9 Some Useful Books
- 9.10 Answer/Hints to Check your Progress Exercises

9.0 OBJECTIVES

After studying this Unit, you should be able to:

- define the concept of the impossible trinity;
- identify the components of the impossible trinity;
- examine the impossibility of achieving all three components simultaneously;
- analyse the empirical evidence in the context of the impossible trinity;
- point out the problem of rupee convertibility in meeting the conditions of the impossible trinity; and
- highlight the importance of the study of the impossible trinity.

9.1 INTRODUCTION

Macroeconomic policy trilemma, also called the impossible trinity, means that a country must choose between free capital mobility, exchange-rate management and an independent monetary policy. Only two of the three are possible. A country that wishes to fix its currency's value and has an interest-rate policy free from outside influence cannot allow capital to flow freely across its borders. The

* Contributed by Prof. S.K. Singh, Retd. Prof. IGNOU

articulators of the concept in the 1960s were *J. Marcus Fleming* and *Robert A. Mundell*. The impossible trinity is a simple rule with deep implications. This trilemma is being faced by many countries. China faced this trilemma when the exchange rate was fixed but the country was open to cross-border capital flows, it cannot have an independent monetary policy. Similarly, Britain has this trilemma when it chose free capital mobility and wanted monetary autonomy it had to allow its currency to float. There are many findings from empirical studies where governments that have tried to pursue all three goals simultaneously have failed. This is the prevailing global monetary scenario where the two-from-three combinations most modern economies choose.

9.2 THE CONCEPT OF IMPOSSIBLE TRINITY

The concept is derived from the academic works of Canadian economist *Robert Mundell* and British economist *Marcus Fleming* in the early 1960s. This principle is frequently called '*the Unholy Trinity*,' the '*Irreconcilable Trinity*,' the '*Inconsistent trinity*' or the Mundell-Fleming '*trilemma*'. This is a concept in international economics which states that it is impossible to have all three of the following at the same time:

- i) a fixed foreign exchange rate;
- ii) free capital movement (absence of capital controls);
- iii) an independent monetary policy.

Under the impossible trinity (a play on '*dilemma*'), two policy positions are possible. If a nation were to adopt position *a*, for example, it would maintain a fixed exchange rate and allow free capital flows, the consequence of which would be a loss of monetary independence. A central bank can only pursue two of the three policies mentioned above simultaneously. To see why consider this example:

Assume that the world interest rate is at 5%. Suppose the home central bank tries to set the domestic interest rate at a rate lower than 5%, for example at 2%. In that case, there will be depreciation pressure on the home currency, because investors would want to sell their low-yielding domestic currency and buy higher-yielding foreign currency. If the central bank also wants to have free capital flows, the only way it could prevent the home currency's depreciation is to sell its foreign currency reserves. Since the foreign currency reserves of a central bank are limited, once the reserves are depleted, the domestic currency will depreciate. Hence, all three of the policy objectives cannot be pursued simultaneously. A central bank has to forgo one of the three objectives. Three out of three are not possible, but two out of three are not bad.

9.3 THEORETICAL UNDERPINNING: MUNDELL – FLEMING MODEL

The *Mundell-Fleming* model integrates international trade and finance into macroeconomic theory. This approach was developed in the early 1960s by the Canadian economist *Robert Mundell* (winner of the 1999 Nobel Prize in economics) and the British economist *J. Marcus Fleming* (1911–1976). Both authors were members of the IMF's research department. They independently extended the traditional Keynesian model to an open economy set-up in which the capital and goods markets are internationally integrated. The resulting research constitutes the original version of the *Mundell-Fleming* model (*Mundell* 1963; *Fleming* 1962). The model states that, in the long run, a central bank that hopes to conduct independent monetary policy must choose between maintaining a fixed foreign exchange rate and allowing the free movement of capital. For instance, a central bank that chooses to increase the total money supply by adopting a loose monetary policy cannot hope to maintain the foreign exchange value of its currency unless it resorts to restricting the sale of domestic currency in the currency market.

Basic set-up*: The model is an extension of the IS-LM model. The traditional IS-LM model deals with the economy under autarky (or a closed economy), whereas the *Mundell-Fleming* model tries to describe an open economy. Typically, the *Mundell-Fleming* model portrays the relationship between the nominal exchange rate and an economy's output (unlike the relationship between interest rate and the output in the IS-LM model) in the short run. The *Mundell-Fleming* model has been used to argue that an economy cannot simultaneously maintain a fixed exchange rate, free capital movement, and an independent monetary policy. The traditional model is based on the following equations.

$$Y = C + I + G + NX \text{ (The IS Curve)}$$

Where Y is GDP,

C is consumption,

I is investment,

G is government spending, and

NX is net exports.

$$\frac{M}{P} = L(i, Y) \text{ (The LM Curve)}$$

Where M is the money supply,

P is the average price,

L is liquidity, i is the interest rate, and Y is GDP

$$BoP = CA + KA \text{ (The BoP Curve (Balance of Payments))}$$

Where CA is the current account and

KA is the capital account

IS components

$$C = C(Y - T, i - E(\pi))$$

Where C is consumption,

Y is GDP,

T is taxes,

i is the interest rate,

$E(\pi)$ is the expected rate of inflation

$$I = I(i - E(\pi), Y - 1)$$

Where I is investment,

i is the interest rate,

$E(\pi)$ is the expected rate of inflation,

$Y - 1$ is GDP in the previous period

$$G = G$$

Where G is government spending, an exogenous variable

$$NX = NX(e, Y, Y^*)$$

Where NX is net exports,

e is the real exchange rate,

Y is GDP,

Y^* is the GDP of a foreign country

BoP components

$$CA = NX$$

Where CA is the current account and

NX is net exports

$$KA = z(i - i^*) + k$$

Where z is the level of capital mobility,

i is the interest rate,

i^* is the foreign interest rate,

k is capital investments not related to i , an exogenous variable

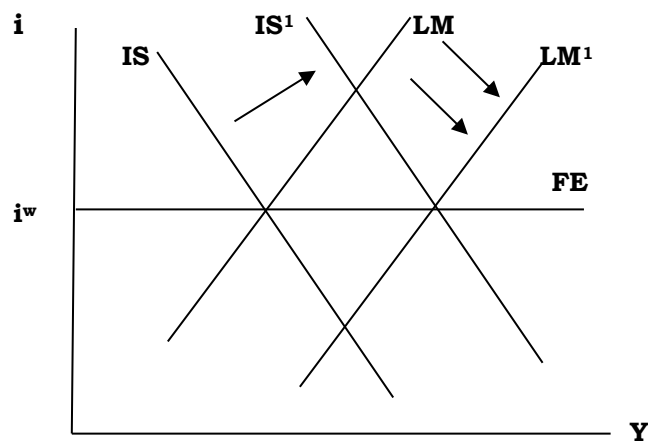
A critical assumption of the model is the equalization of the local interest rate to the global interest rate.

Changes in money supply: An increase in the money supply shifts the LM curve downward. This directly reduces the local interest rate and forces the local interest rate to lower than the global interest rate. This depreciates the exchange rate of local currency through capital outflow. Hot money flows out to take advantage of higher interest rates abroad, so currency depreciates. This leads to the shifting of the IS curve ($Y = C + I + G + NX$) to the right, where the local interest rate equalizes with the global rate. A decrease in the money supply causes the exact opposite of the process.

Changes in government spending: An increase in government expenditure shifts the IS curve to the right. The shift causes the local interest rate to go above the global rate. The increase in local interest causes capital inflow, making the local currency stronger than foreign currencies. A strong exchange rate also makes foreign goods cheaper compared to local goods. This encourages greater import and discourages export and hence lower net export. As a result, the IS returns to its original level, where the local interest rate is equal to the global interest rate. The level of income of the local economy stays the same. The LM curve is not at all affected. A decrease in government expenditure reverses the process.

Changes in global interest rate: An increase in the global interest rate causes upward pressure on the local interest rate. The pressure subsides as the local rate closes in on the global rate. When a positive differential between the global and the local rate occurs, holding the LM curve constant, capital flows out of the local economy. This depreciates the local currency and helps boost net export. Increasing net export shifts the IS to the right. This shift continues to the right until the local interest rate becomes as high as the global rate. A decrease in the global interest rate causes the reverse to occur.

Changes in money supply: Under the fixed exchange rate system, the local central bank only changes the money supply to maintain a specific exchange rate. Suppose there is pressure to depreciate the domestic currency's exchange rate because the supply of domestic currency exceeds its demand in foreign exchange markets. In that case, the local authority buys a domestic currency with foreign currency to decrease the domestic currency's supply in the foreign exchange market. This returns the domestic currency's exchange rate to its original level. Suppose there is pressure to appreciate the domestic currency's exchange rate because the currency's demand exceeds its supply in the foreign exchange market. In that case, the local authority buys a foreign currency with domestic currency to increase the domestic currency's supply in the foreign exchange market. This returns the exchange rate to its original level. A revaluation occurs when there is a permanent increase in the exchange rate and hence, a decrease in the money supply. Devaluation is the exact opposite of revaluation.



An increase in government spending forces the monetary authority to flood the market with local currency to keep the exchange rate unchanged.

Figure 9.1: Changes in IS-LM due to changes in Money supply

The local monetary authority in the framework of a fixed system controls the exchange rate. To maintain the exchange rate and eliminate pressure from it, the monetary authority purchases foreign currencies with local currency until the pressure is gone, i.e., back to the original level. Such action shifts the LM curve in tandem with the direction of the IS shift. This action increases the local currency supply in the market and lowers the exchange rate—or rather returns the rate to its original state. In the end, the exchange rate stays the same but the general income in the economy increases.

If the global interest rate increases above the domestic rate, capital flows out to take advantage of this opportunity. Hot money flows out of the economy. This would depreciate the home currency, so the central bank may buy the home currency and sell some of its foreign currency reserves to offset this outflow. This decrease in the money supply shifts the LM curve to the left until the domestic interest rate is the global interest rate. The opposite occurs if the global interest rate declines below the domestic rate. Hot money flows in, and the home currency appreciates. The central bank offsets this by increasing the money supply (selling domestic currency, buying foreign currency), the LM curve shifts to the right, and the domestic interest rate becomes the global interest rate.

Differences from IS-LM: It is worth noting that some of the results from this model differ from the IS-LM because of the open economy assumption. The result for a large open economy on the other hand falls within the result predicted by the IS-LM and the *Mundell-Fleming* models. The reason for such a result is that a large open economy has both the characteristics of autarky and a small open economy. In the IS-LM, the interest rate is the key component in making both the money market and the goods market in equilibrium. Under the *Mundell-Fleming* framework of a small open economy, the interest rate is fixed and equilibrium in both markets can only be achieved by a change in the nominal exchange rate.

Example: A much-simplified version of the *Mundell-Fleming* model can be illustrated by a small open economy, in which the domestic interest rate is exogenously predetermined by the *world interest rate* ($r=r^*$). Consider an exogenous increase in government expenditure, the IS curve shifts upward, with LM curve intact, causing the interest rate and the output to rise (partial crowding out effect) under the IS-LM model. Nevertheless, as the interest rate is predetermined in a small open economy, the LM¹ curve (of exchange rate and output) is vertical, which means there is exactly one output that can make the money market in the equilibrium under that interest rate. Even though the IS¹ curve can still shift up, it causes a higher exchange rate and the same level of output (complete crowding out effect, which is different in the IS-LM model). This example makes an implicit assumption of a flexible exchange rate. The *Mundell-Fleming* model can have completely different implications under different exchange rate regimes. For instance, monetary policy becomes ineffective under a fixed exchange rate system, with perfect capital mobility. An expansionary monetary policy resulting in an outward shift of the LM curve would in turn make capital flow out of the economy. The central bank under a fixed exchange rate system would have to intervene by selling foreign money in exchange for domestic money to depreciate the foreign currency and appreciate the domestic currency. Selling foreign money and receiving domestic money would reduce real balances in the economy, until the LM curve shifts back to the left, and the interest rates come back to the world rate of interest i^* .

*This section is largely based on the contributions of over twenty authors

(<http://gobernanzamedioambiental.org/brugger/Mundell%E2%80%93Fleming%20model.pdf>)

There is another way to understand the trilemma. Let us look at the diagram given below:

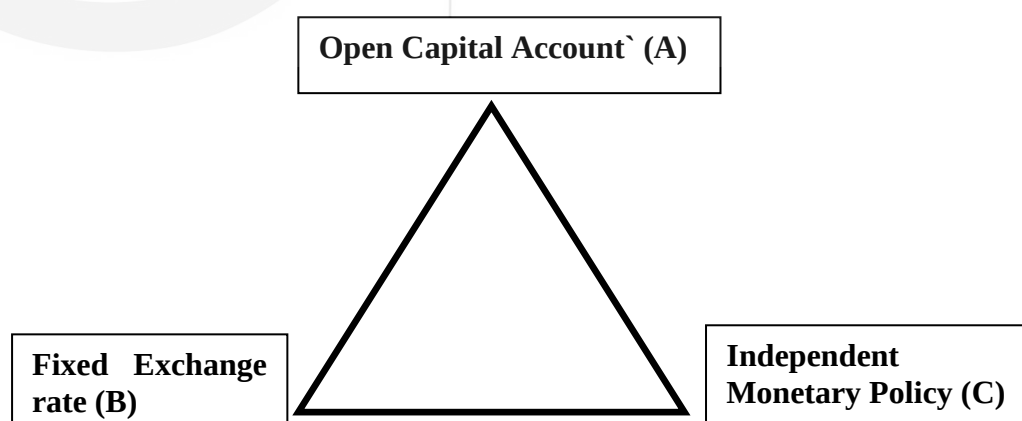


Figure 9.2: The “Impossible trinity” of Capital, Money and exchange

The diagram represents the *impossible trinity*. In this schematic, the open capital account is labelled "A", the fixed exchange rate is labelled "B," and the independent monetary policy is labelled "C."

In terms of the Figure 9.2 (Oxelheim, 1990), the following are the options:

- i) “A”: A stable exchange rate and free capital flows (but not an independent monetary policy because setting a domestic interest rate that is different from the world interest rate would undermine a stable exchange rate due to appreciation or depreciation pressure on the domestic currency).
- ii) “B”: An independent monetary policy and free capital flows (but not a stable exchange rate).
- iii) “C”: A stable exchange rate and independent monetary policy (but no free capital flows, which would require the use of capital controls).

The theory of the *impossible trinity* is that a country can't achieve A+B+C at the same time. Any country that attempts this is doomed to fail. That's the crux of the theory. *Mundell* was of the view that the effects of monetary and fiscal policy in an open economy depend on capital mobility. In particular, he has highlighted the importance of exchange rate systems. Under floating exchange rates, monetary policy is a powerful tool; fiscal policy is powerless. Under fixed exchange rates, fiscal policy is effective.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) What do you mean by impossible trinity?

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- 2) Describe the main components of the *Mundell-Fleming* model.

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- 3) Explain the process involved in the *Mundell-Fleming* model.

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9.4 IMPOSSIBLE TRINITY: ALTERNATIVE SCENARIOS COUNTRIES' EXPERIENCE

The choices made by policymakers are not so clear-cut. The concept of the impossible trinity went from theoretical curiosity to becoming the foundation of open economy macroeconomics in the 1980s, by which time capital controls had broken down in many countries, and conflicts were visible between pegged exchange rates and monetary policy autonomy. While one version of the impossible trinity is focused on the extreme case – with a perfectly fixed exchange rate and a perfectly open capital account, a country has absolutely no autonomous monetary policy; the real world has thrown up repeated examples where capital controls are loosened, resulting in greater exchange rate rigidity and less monetary-policy autonomy.

Europe's *exchange rate mechanism* (ERM) was born after the death of the Bretton Woods System. No member violated the impossible trinity. The German unification in 1989 was the first big shock. The second big shock was the single market. The single market required capital mobility and free movement of capital. With fixed exchange rates and no control over capital, member nations had to implicitly give up their autonomy over monetary policy. Since Germany was the biggest economy, they aligned themselves with the policies of Germany's Bundes Bank. But the reunification called for a very tight monetary policy in Germany and, therefore, in other member-nations. Britain needed something loose and different. Britain walked out of the ERM because it knew that the impossible trinity could not be violated. Britain regained interest rate autonomy, retained capital mobility and regained floating exchange rates. This explains why Britain is a non-euro country.

In practice, many emerging markets are fearful of letting the exchange rate move sharply, so they choose to sacrifice either free capital mobility (by introducing capital controls, or by adding to or depleting their foreign-currency reserves) or monetary independence, by giving priority to currency stability over other targets. China for instance is currently up against the classic trilemma involving the impossible trio. The country wants to keep the *Yuan* stable because it aspires to make it an international reserve currency akin to the US dollar. So the currency is allowed to fluctuate only 2 per cent on either side of a peg determined by the Chinese central bank. China is also trying to accelerate the move towards making the *Yuan* freely convertible as that is one of the prerequisites for an international currency. Attempting to sustain the current rate of exchange will lead to the depletion of forex reserves. This is impeding China from following the independent monetary policy needed to kick-start growth. China wants eventually to liberalise its capital account as a stepping stone to a modern financial system. To do so, it will have to live with a volatile *yuan*.

With open capital markets and a currency board, Hong Kong implicitly accepts foreign monetary policy, which can sometimes be too tight or too easy. The fixed

exchange rate provides a stable environment for importers and exporters. Countries such as Australia and New Zealand have chosen domestic price stability in the form of an inflation target, and accept a high degree of exchange rate volatility. India has a multiple indicators approach and looks at everything. Surging capital flows is a problem in most countries. Countries like Korea, NZ and the Philippines have inflation targets and capital flows have been a problem with them. Ideally, there should have seen some appreciation in these countries. However, the currency has appreciated only in India, Korea and China from 1994 levels. The consensus is to have free capital inflows, have a stable monetary policy and let the currency float.

India has been able to actively manage the exchange rate and limit exchange rate volatility relative to other emerging market economies, by building up international reserves and intervening actively in the foreign exchange market. Such reserve management has also helped to some extent in regaining control over monetary policy even in the face of capital inflows. However, the steps RBI took in 2013 to defend the currency have resulted in a reversal of the direction of interest rates in the economy. The then RBI governor *D. Subbarao*, in his statement on the review of the monetary policy on 30 July 2013 noted that India is currently caught in a classic '*impossible trinity*'. Put differently, to protect the rupee, RBI has lost some independence in how it makes the monetary policy.

In sum, in the modern world, given the growth of trade in goods and services along with fast paced of financial innovation and lacking effective control over the free movement of capital, the impossible trinity asserts that a country has to choose between reducing currency volatility and running a stabilising monetary policy. It cannot do both. The combination of the three policies is known to cause the financial crisis. The Mexican peso crisis (1994–1995), the 1997 Asian financial crisis (1997–1998), and the Argentinean financial collapse (2001–2002) are often cited as examples. In particular, the East Asian crisis (1997–1998) is widely known as a large-scale financial crisis caused by the combination of the three policies which violate the impossible trinity. *Mundell* always says to let capital flow and defend exchange rates. *Hélène Rey* has observed that “for the past few decades, international macroeconomics has postulated the '*trilemma*': with free capital mobility, independent monetary policies are feasible if and only if exchange rates are floating. The global financial cycle transforms the trilemma into a '*dilemma*' or an '*irreconcilable duo*': independent monetary policies are possible if and only if the capital account is managed”.

In this context, it is well to remember what James Rickards has emphasized; there is one big exception to the impossible trinity. That's the United States. The U.S. sets interest rates independently and has an open capital account. The U.S. does not officially peg the dollar to any other currency (thus technically not breaking the Impossible Trinity). But it does work with other countries to allow them to informally peg to the dollar. Why does the U.S. not suffer adverse

consequences? Because the U.S. does not need foreign exchange. The dollar is the leading reserve currency in the world (about 60% of global reserves and about 80% of global payments), so the U.S. can never run out of foreign exchange to pay for things, it can just print more dollars! This is what the French called the dollar's 'Exorbitant Privilege' in the 1960s.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Why is it difficult for countries to circumvent the impossible trinity?

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2) Discuss the policies adopted by European countries

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3) Give examples of financial crises because of the non-observance of the impossible trinity.

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9.5 IMPORTANCE OF IMPOSSIBLE TRINITY

The model serves as an introduction to the formal theory of policymaking. This model cannot be relegated to textbooks only. It affects the policies of countries and central banks across the globe. This concept is impossible to ignore, especially if one is a keen follower of macros. Theoretically, policy-makers should manage macroeconomic policies following the rule that '*for each goal might correspond only one macroeconomic policy instrument*'. This means that even a floating exchange rate regime does not release policy-makers from facing the '*trilemma*' of economic policy. No government can achieve more targets than it has instruments. The *Mundell* model implied an optimal assignment and, more important, was drawn up within the *Tinbergen* target-instrument framework. Many of the more sophisticated models followed have continued in this tradition.

In this regard, the *Mundell-Fleming* model has been successful in the theoretical predictions in matching empirical facts such as the effects of U.S. macroeconomic policies in the 1980s and helping explain the model's influence among academics and policymakers. *Mundell's* contribution to international macroeconomics is extraordinary. The '*impossible trinity*' is among his best. It is

axiomatic. Understanding and using the *impossible trinity* works wonders if one can spot the right conditions and set up the trades in advance of the inevitable policy failures of the central banks. Those failures (which do happen) represent some of the best profit-making opportunities of all. While citing the example of England, *James Rickards* has observed how by understanding the *impossible trinity* *George Soros* broke the Bank of England on September 16, 1992 (still referred to as ‘*Black Wednesday*’ in British banking circles). *Soros* also made over \$1 billion that day. The importance of the concept may be recapitulated as under:

- i) The impossible trinity is a tool to separate countries with good policies from those with bad policies. It is also a tool used to make accurate forecasts based on the sustainability of those policies.
- ii) Using the impossible trinity to spot trends and make investment decisions requires a lot of analysis. There are three policies that can go wrong (A, B, and C above), and three ways to fix them (again, A, B, and C). That gives nine basic scenarios ($3 \times 3 = 9$), and many more when one considers hybrid or combined policies, (a country could devalue its currency and slap on capital controls at the same time). There are always expert practitioners on the lookout for non-sustainable policies that are bound to break. This is a kind of strategic intelligence.
- iii) The policy challenges associated with the ‘*impossible trinity*’ have become more relevant to developing countries as more and more of them have taken steps to liberalize the capital account of their balance of payments to permit freer flows. Over the years, several countries have mishandled this trilemma. It is important to remember that domestic policy-makers should not deviate from this discipline from time to time.
- iv) Countries must keep in mind the key lessons to be drawn from the experience of other countries. Flouting the basic premises of the *Mundell – Fleming model* (the ‘*impossible trinity*’) has inevitable consequences. It leads to unsustainable macroeconomic imbalances. Failure to take early remedial action would result in a steady deterioration of macroeconomic fundamentals. The earlier the remedial measures are implemented the less painful will be the adjustment. The painful corrective austerity measures have eventual political costs, particularly if stabilization measures are implemented without supporting structural reforms which strengthen the growth framework of the economy.
- v) Impossible trinity has highlighted the main challenges that emerging economies face today as they integrate with global capital markets in managing the trade-offs between the trinity. The rise in the volatility of global capital flows has made macroeconomic management increasingly complex, especially for emerging economies that are striving to achieve a balance between the diverse objectives of robust growth rate, sustainable current

account deficit, competitive exchange rate, access to adequate external capital for financing investment, moderate inflation, and sufficient international reserves. This has also reignited the debate on suitable macroeconomic management measures. *Sooner or later bad economics becomes bad politics.* The authorities must move beyond politics to embark upon sound economic policy-making which recognises the discipline imposed by the 'impossible trinity'.

9.6 IMPOSSIBLE TRINITY AND DEMAND FOR CAPITAL ACCOUNT CONVERTIBILITY (CAC) OF INDIA'S RUPEE

The move towards making the Indian rupee fully convertible on the capital account has attracted considerable debate. The critics have claimed that full convertibility or capital mobility and freely floating exchange rates are inimical to India's interests. The critics have referred to Britain's crisis in 1992 and the Asian crisis in 1997 to buttress their criticism. However, the criticisms have been contested by the advocates of capital account convertibility. Advocates have argued that capital mobility and freely floating exchange rates are not joint policy issues. One can exist without the other. The crisis in Britain erupted when policymakers understood the axiomatic inviolability of the impossible trinity. The Asian crisis was craftily brewed by central banks that dodged the impossible trinity.

In a managed float economy, the rupee floats somewhat. Although the rupee is not pegged to any currency, India does not allow free movement on the capital account. Capital control is put in place to preserve financial stability as large inflows and outflows can lead to financial instability. The RBI is not defending the rupee by selling foreign currency in the market, but if it did, the impact on the interest rate would have been the same. Selling foreign exchange would reduce the supply of rupees in the market which would have resulted in higher rates. But the RBI has made it clear that it would like to move towards full convertibility over the long term. If that happens, RBI may not be able to maintain the tight reign on interest rates. An independent monetary policy will be difficult given that the rupee is nowhere close to becoming a reserve currency like the dollar.

The researchers have observed that India's capital account liberalization that began in the early 1990s has since then been a steady albeit slow and gradual process. Regarding opening up the financial markets to foreign investment, Indian authorities have always proceeded with a lot of caution and apparent scepticism about the benefits of foreign capital balanced against the potential instability that such capital inflows might trigger. On one hand foreign capital offers a diversification of opportunities to investors and provides avenues to bridge the gap between domestic saving and investment, and on the other unbridled flows can fuel inflationary pressures, fan asset price bubbles, and

sudden reversals in capital inflows can lead to instabilities and even crises in financial and currency markets. The decade of the 1990s was replete with several such incidents of financial crises all over the emerging world as seen in Mexico (1994), East Asia (1997), Russia (1998), Brazil (1999) and eventually Argentina (2000). Rampant volatile debt inflows in the domestic economies triggered all these episodes of crises. (Reddy, 2008).

Though India is now more open in absolute terms than it used to be during the early 1990s, the cautious and gradual approach towards capital account liberalization has meant that India has significantly lagged behind other countries. Even today, more than two decades after India started opening up its capital account, there exists an extensive array of restrictive controls imposed by authorities on different categories of foreign investment to actively manage the capital account. India's capital account seems to be only partially open and the exchange rate is in a managed floating regime. India has also been accumulating large amounts of foreign currency reserves since the early 2000s.

India does not allow free movement on the capital account. Capital control is put in place to preserve financial stability as large inflows and outflows can lead to financial instability. Issues related to the capital account constitute a subject upon which divergent views have existed and a professional consensus is yet to emerge. CAC is not a standalone objective but a means for higher and stable growth. There is no need or use to hustling towards CACs; the movement should be at a pace the country is comfortable with, considering all relevant factors and the environment, in which the country has to function.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What are the implications of the impossible trinity in policymaking?

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2) Why do we study the impossible trinity? Point out its practical significance.

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3) What is capital account convertibility (CAC)? How are capital A/c convertibility and current A/c convertibility different?

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9.7 LET US SUM UP

After completion of this unit, you will understand that, the impossible trinity also called the Mundell-Fleming trilemma or simply the trilemma expresses the limited options available to countries in setting monetary policy. According to this theory, a country cannot simultaneously achieve the free flow of capital, a fixed exchange rate and independent monetary policy. Pursuing any two of these options necessarily closes off the third.

The '*impossible trinity*' is a fundamental contribution of the Mundell-Fleming framework. Confronted with this Trilemma policymakers have to choose a combination of any two objectives because all three goals can be mutually inconsistent.

Today most countries allow their currencies to float. That way, the currency adjusts to the waxing and waning of capital flows, allowing interest rates to respond to the domestic business cycle. The trilemma is not as simple as it appears since most countries lack monetary policy independence whether or not they have free exchange rates and capital flows. The reason has to do with the overwhelming influence of the Federal Reserve.

With autonomous monetary policy, the exchange rate is on autopilot. With fixed exchange rates, the monetary policy is on autopilot. This is what makes the entire learning so good. One not only knows what the impossible trinity is, but one also knows which country is doing what.

India adopted a calibrated and hierarchical approach towards capital account liberalization, prioritizing some flows over others. CAC need not be India's overriding priority.

9.8 KEY WORDS

Trade-offs	Something is being given up for something else.
Asian financial crisis	The Asian Financial Crisis of 1997 (also called the 'Asian Contagion') was a financial crisis that affected many Asian countries, including South Korea, Thailand, Malaysia, Indonesia, Singapore and the Philippines. After posting some of the most impressive growth rates in the world at the time, the so-called 'tiger economies' saw their stock markets and currencies lose about 70% of their value.
Tinbergen target-instrument framework	According to the well-accepted 'Tinbergen Rule' a policy instrument can be effective only if it has a single objective. No government can achieve more targets than it has instruments.

For example, when there are two targets, two policy instruments are required to achieve both objectives. The interest rate is better suited to target domestic imbalances. Targeting exchange rates through market intervention is better suited to targeting external imbalances. The Tinbergen Rule has been used to criticise multi-target policy instruments for being inefficient.

Macroeconomics Management Management of issues relating to GDP, fiscal policy, monetary policy, trade policy, growth strategies, sovereign wealth funds, etc.

**Capital account
convertibility (CAC)**

Is a feature of a nation's financial regime that centres on the ability to conduct transactions of local financial assets into foreign financial assets freely or at country-determined exchange rates.

Financial Integration

There is no universal definition of 'financial integration'. Nevertheless, the term typically encompasses concepts like financial openness, free movement of capital and integration of financial services. The Financial System Inquiry of 2014 considered financial integration to be a country's 'financial connectedness' with other countries, noting that greater financial integration tends to increase capital flows and equalise prices and returns on traded financial assets in different countries.

International Reserves

International reserves are any kind of reserve funds that can be passed between the central banks of different countries. International reserves are an acceptable form of payment between these banks. The reserves themselves can either be gold or else a specific currency, such as the dollar or euro.

Capital control

Capital control represents any measure taken by a government, central bank or other regulatory body to limit the flow of foreign capital in and out of the domestic economy. These controls include taxes, tariffs, outright legislation and volume restrictions, as well as market-based forces. Capital controls can affect many asset

classes such as equities, bonds and foreign exchange trades.

Capital movement

The transfer of capital between countries. It refers to the movement of money for investment, trade or business production, including the flow of capital within corporations in the form of investment capital, capital spending on operations and research and development (R&D).

Sterilized Intervention

Is the purchase or sale of foreign currency by a central bank to influence the exchange value of the domestic currency, without changing the monetary base. Empirical evidence suggests that sterilized intervention is generally incapable of altering exchange rates.

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9.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check your progress 1

- 1) See section 9.2
- 2) See section 9.3
- 3) See section 9.3

Check your progress 2

- 1) See section 9.4
- 2) See section 9.4
- 3) See section 9.4

Check your progress 3

- 1) See section 9.5
- 2) See section 9.5
- 3) See section 9.6

UNIT 10 APPROACHES TO BALANCE OF PAYMENTS *

Structure

10.0 Objectives

10.1 Introduction

10.2 Approaches to BoP

10.2.1 Elasticity approach

10.2.2 The absorption approach

10.2.3 Keynesian Approach

10.2.4 The monetary approach

10.2.4.1 Tenets of the monetary approach

10.2.4.2 Monetary approach and fixed exchange rate

10.2.4.3 Monetary approach and flexible exchange rate

10.2.4.4 Criticisms of the monetary approach to the balance of payments

10.2.5 Synthesising all the approaches

10.3 Let us sum up

10.4 Key Words

10.5 Some Useful Books and References

10.6 Answer/Hints to Check your Progress Exercises

10.0 OBJECTIVES

After studying this Unit, you should be able to:

- identify different approaches to BoP;
- explain the main ingredients of these approaches;
- distinguish between monetary approach under fixed and flexible exchange rates;
- point out the main criticisms levelled against the monetary approach;
- make a synthesis of different approaches; and
- highlight the importance of these approaches in making BoP policy.

* Contributed by Prof. S.K. Singh, Retd. Prof. IGNOU

10.1 INTRODUCTION

Some of the problems which policymakers generally face while analyzing the balance of payments are related to the disequilibrium deficit or surplus and their causes. Disequilibrium in the balance of payments is a general economic phenomenon in most developing countries. In this context, it is important to know the role of a country's central bank and the policy options it has for maintaining equilibrium in the balance of payments. Many theories in the field of international economics explain the balance of payments disequilibrium and the movements of international reserves in one way or another. Mainly there are four approaches. In the elasticities and absorption approaches, the focus is on the trade balance with unemployed resources. These approaches are associated with *Keynesian* theory. The distinctive feature of the monetary approach to the balance of payments is that it examines the implications for the stock equilibrium of the continuing flow of financial assets required to finance a continuing balance of payments surplus or deficit. A central proposition of the monetary approach to the BoP is that the balance of payments is a monetary phenomenon requiring analysis with the tools of monetary theory. This unit reviews the major approaches to BoP adjustments.

10.2 APPROACHES TO BoP

The alternative approaches to BoP adjustment are explained in this section. These approaches are as under:

- i) The Elasticity Approach;
- ii) The Absorption Approach;
- iii) The Keynesian approach; and
- iv) The monetary approach.

Let us discuss all the approaches one by one.

10.2.1 Elasticity approach

We begin first by considering the elasticity approach. The elasticities approach is based on the premise that relative prices determine exports and imports. In this way elasticities approach argues that the number of exports and imports will depend upon domestic prices, foreign prices and the exchange rate. This approach applies the *Marshallian* analysis of elasticities of supply and demand for individual commodities to the analysis of exports and imports as a whole. It is generally assumed that exports depend on the price of exports, and imports depend on the price of imports. These relations are then translated into elasticities.

The effect of devaluation on the trade balance depends on four elasticities:

- a) the foreign elasticity of demand for exports;
- b) the home elasticity of supply;
- c) the foreign elasticity of supply of imports; and
- d) the home elasticity of demand for imports.

The immediate effect of devaluation is a change in relative prices. If a country devalues by, for example, 20 percent, it means that import prices increase by 20 percent counted in home prices. An increase in import prices leads to a fall in the demand for imports. At the same time, import-competing industries will be in a better competitive situation. Exporters will receive 20 percent more in the home currency for every unit of foreign currency they earn. They can, therefore, lower their prices counted in foreign currency and will become more competitive. How much they can expand sales abroad depends primarily on the foreign demand elasticities for their goods.

The larger the elasticity, the greater the fall in the volume of imports. The value of the demand elasticity of imports depends, of course, on what types of goods the devaluing country imports. When the exporters, because of devaluation receive more for every unit of foreign currency they earn, they can lower their prices quoted in foreign currency. When they lower their prices they should be able to sell more. How much the quantity exported increased depends on the demand elasticity confronting the country's exporters.

The essence of the theory is embodied in the famous **Marshall Lerner condition**: an improvement in competitiveness will improve the balance of payments if and only if the sum of the price elasticities of demand for imports by residents and exports by non-residents exceeds unity. The crucial feature of the condition is that it is the *sum* of the elasticities which must exceed one, not each elasticity separately. Hence both exports and imports can be price inelastic but for the condition to be satisfied. In effect, the exchange rate clears the balance of payments. A criterion for a change of the balance of trade in the desired direction can be established, assuming that export and import prices adjust to equate the demand for and supply of exports and imports.

The effect on the price level depends primarily on the economic policy accompanying devaluation. The inflationary impact could be limited if a tight monetary and fiscal policy is pursued jointly with devaluation. Another consideration to take into account is the effect of devaluation on income distribution. Again devaluation should result in a reallocation of resources away from the sector producing non-traded goods and into exports and import-competing sectors. In general, we can say that the factors of production employed in the export and import-competing sectors will benefit from devaluation.

The *Marshall-Lerner condition* is built on some drastic simplifications. It roughly assumes that the supply elasticities are large and that the trade balance is in equilibrium when devaluation takes place. The sort of mechanism ignored by Marshall-Lerner includes such processes as:

- i) a change in relative prices increases exports;
- ii) the increase in exports generates an increase in income;
- iii) the higher level of income leads to more imports.

This latter increase in imports can be such that there is no improvement in the balance of payments despite the rise in exports. However, none of these assumptions invalidates the spirit of the *Marshall-Lerner* condition, which says that the larger the respective demand elasticities, the more favourable the effect of devaluation on the trade balance. So long as the sum of elasticities exceeds unity, then an improvement in competitiveness will improve the balance of payments i.e., the *Marshall Lerner* condition.

The significance of Marshall-Lerner condition: Opinion is divided about the relevance of Marshall-Lerner, or competitiveness, to the balance of payments. On the one hand, some economists believe that competitiveness is a major determinant of the balance of payments. On the other hand, according to *Johnson* “it should be emphasized that the analysis of the effects of devaluation is completely independent of any critical magnitude condition applying to the elasticities of international demand. The relevant stability condition is monetary theoretic. The familiar elasticity condition (sum-of-the-elasticities-of demand—greater-than-unity) is completely irrelevant”.

However, it should be noted that the elasticities approach in general can be relevant to the analysis of inflation and unemployment. The income effect, whose size does depend in part on *Marshall-Lerner* analysis, is not an unwelcome by-product but the object of the policy. Similarly, exogenous change in competitiveness e.g. by income policy, as a means to reduce unemployment is important in policymaking. The major flaw with the elasticities approach is that it only looked at the current account balance, not the overall balance of payments.

10.2.2 The Absorption Approach

Sydney Alexander of the IMF first presented the absorption approach in 1952. He sought to look at the balance of trade from the point of view of national income accounting:

Y = domestic production of goods and services

E = domestic absorption of goods and services, or domestic total expenditure

BT = balance of trade

$BT = Y - E$

The above identity is useful in pointing out that an improvement in the balance of trade calls for an increase in production relative to absorption. The absorption approach can be said to work only in the presence of unemployed resources. This approach argued that questions regarding the effects of changes in economic variables on the balance of payments were best assessed by ascertaining the effects on output and absorption. It concentrates on the relationships between real expenditure and real income and the relationships between these price levels. Absorption here is the name given to the aggregate of domestic demand ($C + I + G$), which is the number of goods and services taken off the market domestically. To *Alexander*, the foreign balance '**B**' is the difference between the total output of goods and services '**Y**' and the total absorption '**A**' of these goods and services by the home economy. Thus, $B = Y - A$.

If the total output is larger than the total expenditure, the country will have a surplus in its BoP and if the total expenditure is larger than the total output the country will have a deficit, and if output equals expenditure, the BoP will be in equilibrium. If a country has a deficit it can, in principle, close the deficit in one of two ways; by reducing expenditure or by increasing output. It is often difficult to increase output in the short run especially if the country already has full employment. Therefore, the chief means for reducing a deficit is usually an expenditure-reducing or switching policy. A tighter *monetary policy and fiscal policy* can be viewed as an efficient means of implementing an expenditure-reducing policy. If the country has a deficit in the BoP, it can pursue a tighter monetary and fiscal policy. There will be a deflationary effect on the national income and lead to a fall in imports, or at last act as a brake on the increase in imports. It will also have a positive effect on exports and import-competing industries.

The criticisms against the absorption approach: The following are some of the weaknesses of this approach:

- i) *Machlup* has observed that certain elasticity considerations in the model had been ignored. *Alexander* had seen these situations in terms of different supply and demand conditions for commodities, which necessarily involved elasticities. However, *Alexander's* analysis played down the role of elasticity in his model.
- ii) The estimation required to bring the absorption approach up to the '*operational*' level was just as formidable as those involved in the elasticities approach. In the case of the absorption approach the relevant propensities had to be known before it was possible to say whether devaluation would be beneficial to the trade balance.
- iii) It was necessary to embody within the model the fact that an improvement in the balance of trade of a devaluing country involved opposite changes in the trade balance of the rest of the world.

- iv) This approach suffered from the difficulty that it was not suited for examining changes that only indirectly affected output and absorption: changes such as devaluation or inflation.
- v) An important flaw and one shared with the elasticities approach is that it only looked at the current account balance and not at the overall balance of payments.

The result of these criticisms of the absorption approach was a reformulation which included elasticity effects and was more nearly a synthesis of the elasticity and the income approaches. It is important to realise how neatly the absorption model complements the elasticities approach. The traditional approach ignored supply-side effects and income effects. The absorption approach looks only at these two effects. The two approaches can be combined. *In brief*, the absorption approach focuses on the key factors omitted from the traditional approach and provides a general framework of analysis. Its single most virtue stems from its very existence, the idea that the balance of payments is a macroeconomic variable.

10.2.3 Keynesian Approach

The *Keynesian* balance of payments analysis is a coherent presentation of the macroeconomic case for import controls and devaluation. The *Keynesian* balance of payments theory was the absorption model of Alexander, of the IMF, in the early 1950s. As in all *Keynesian* models, the balance of payments on the current account, is analysed as a macroeconomic phenomenon in the goods market. The current balance of payments will necessarily equal the difference between aggregate domestic output and aggregate domestic expenditure (with a surplus if the output is large and vice versa). This conclusion follows from a manipulation of the basic national income; output ' O ' and expenditure ' E ', where $O=E$.

Although the part played by income changes in BoP adjustment is Keynesian in approach and method, Keynes himself took no direct part in its formulation or development. The Keynesian approach made possible the formulation of a new approach to the adjustment process by *Mrs Joan Robinson, R.F. Harrod, Fritz Machlup and others*. This new approach showed that an imbalance in BoP involved an adjustment in income, employment and output irrespective of what changes took place in prices and of how the deficit was financed. It showed an interaction between the BoP and national income which was a dual one: the '*adjustment process*', by which a BoP is adjusted by changes in the income levels of the home country and the rest of the world; and the '*transmission process*', which shows how variations in the national income of one country may through their balances of payments cause variations in the national incomes of other countries.

Thus, with the development of the income approach to BoP adjustment, a marriage was effected between the theory of International trade and business cycle theory from which much could be expected. It explains not only the adjustment process but also the transmission process. It shows that since deficits are adjustable by relative movements of national income, by reduction of income in the deficit country and increase of income in the surplus country, correction of a deficit must under static productivity conditions, necessarily involve a reduction in national income. The full condition of equilibrium for a country was now redefined as one in which total external receipts equalled total external payments at full employment. This linkage of external equilibrium for a country with its level of income and employment by the new approach enabled a much clearer and most realistic view to be taken of BoP policy in modern times.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What is meant by the Marshall-Lerner condition?

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2) Explain the main features of the absorption approach.

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3) The main merit of the Keynesian approach lies in integrating national income with BoP.

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4) What are the distinctive features of elasticity, absorption, and the Keynesian approach to BoP?

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5) How do these three approaches differ?

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10.2.4 The Monetary Approach

In this section, we examine the monetary approach to the balance of payments.

Evolution of monetary approach: *Alan Barrett* is of the view that some controversy surrounds the early origins of the monetary approach to BoP. Essentially the controversy is over whether *David Hume* or *Richard Cantillon* is the true forefather of the approach. *Ludwig von Mises* can be considered a forerunner of the monetary approach because he analysed the causes and cures of a persistent balance-of-payments disequilibrium. *Mises's* contribution to balance-of-payments analysis should be hailed as a doctrinal milestone in the development of the monetary approach and, much more importantly, as a shining exemplar of methodological individualism in monetary theory.

The monetary approach was developed in the 1950s and 1960s by the International Monetary Fund's research department under *Jacques J. Polak*, and by *Harry G. Johnson*, *Robert A. Mundell*, and their students at the University of Chicago and became fully developed during the 1970s. The monetary approach represents an extension of domestic monetarism (stemming from the Chicago school) to the international economy in that it views the balance of payments as an *essentially monetary phenomenon*. That is, money plays a crucial role in payments.

Mundell included the capital account of BoP in the discussion, and incorporated two policy instruments. One was the growing reluctance of countries to use devaluation as a policy instrument, and the other was the growing constraints on the use of trade barriers arising from G.A.T.T negotiations. While it was also *Mundell* who recognized that, in a model of capital mobility, the central bank controls not the money supply and employment but domestic credit and the balance of payments, *Harry Johnson* popularized the concept of the fundamentally monetary nature of the balance of payments. *Jacques Polak's* approach to monetary management in the context of the balance of payments disequilibrium evolved in the 1950s in work concerned with Latin American economies. The monetary approach to the balance of payments is an extension of closed economy monetary theory, stressing the stability of the demand for money function and considering the channels through which changes in the money supply, out of line with changes in money demand, affect the economy.

10.2.4.1 Tenets of monetary approach

To start with let us understand the basic assumptions of this approach which are as under:

- i) Assumes flexible exchange rates. In other words, the currencies in question are traded in foreign exchange markets with minimal or no government intervention.
- ii) It focuses exclusively on the behaviour of international investors trying to decide between securities denominated in two different currencies. Based on

this the monetary approach is also called the *asset approach to exchange rate determination*.

- iii) The approach emphasizes the relevance of the changes in monetary policy and the resulting changes in real returns on securities denominated in different currencies. After investors observe these changes in real returns, they express their preference for security, which leads to buying or selling certain currencies and, therefore, changes in the exchange rate.
- iv) The demand for money is a stock demand and is a stable function of income, prices, wealth and interest rate.
- v) The supply of money is a multiple of the monetary base, including domestic credit and the country's foreign exchange reserves.
- vi) The demand for nominal money balances is a positive function of nominal income.

Given these assumptions, the BoP is regarded as a monetary phenomenon. The very concept of a BoP implies the existence of money. The proponents of the monetary approach regard the balance of payments as being fundamentally a monetary phenomenon. An analysis which does not place money at the centre of attention is seen as inferior. For them, the attraction of the monetary approach is that it analyses the relationships directly relevant to the money account and not in terms of other behavioural relationships which are only indirectly relevant to the money account. Considerations of this relationship can lead us to the conclusion of the monetary approach to explaining the relationship among monetary policy, domestic credit policy, and the flow of international reserves in the context of BoP. Most components of the balance of payments are directly or indirectly affected by a country's domestic credit and monetary policy. The importance of monetary policy is to restore equilibrium in the money market. Some of the macro-variables such as nominal interest rate, GDP and inflation are affected by contractionary/expansionary monetary policy. And, thus, it has a significant effect on domestic credit as well as the balance of payments of a country.

This is based on the simple idea that the balance of payment surplus or deficit is caused by disequilibrium in the money market. A BoP deficit or surplus represents a transient stock-adjustment process evoked by an initial inequality between actual and desired money stocks. The monetary approach to the BoP assumes fixed exchange rates. Whenever the economy has disequilibrium in the money market, it will result in a payments imbalance but this disequilibrium will be self-correcting through time and there is no need for correcting the disequilibrium through policy adjustments. The main reason for the self-correcting mechanism is that the stable demand function for money relates to money as a stock, not as a flow. When the desired stock is reached, the inflow or outflow of foreign funds stops and so does the balance of payment surplus or deficit. According to *Johnson*, the central point of the monetary approach to BoP

is that balance-of-payments deficits or surpluses reflect stock disequilibrium between demand and supply in the market for the money.

Although real factors are not excluded, the demands for the supply of money are the central theoretical relationships used in the analysis of the balance of payments. An increase in domestic credit raises money supply relative to money demand, other things being equal. So the balance of payments must go into deficit to reduce the money supply and restore money market equilibrium. This approach is an explanation of the overall balance of payments. It explains changes in the balance of payments in terms of the demand for and supply of money. According to this approach, *a balance of payments deficit is always and everywhere a monetary phenomenon*. Therefore, it can only be corrected by monetary measures.

10.2.4.2 Monetary approach and fixed exchange rate

The monetary approach implies that a country has no control over its money supply under fixed exchange rates. The money supply is endogenous rather than exogenous or a policy variable. The pursuit of any domestic objective, such as price stability, by altering the domestic component of the monetary base will be frustrated by offsetting changes in the international component through reserve flows. If a country cannot pursue sterilization policies, all the government can do is control the composition of the money supply, not its level. This implies that under the situation the government cannot determine the level of the monetary base and hence money supply, but only the division of the base into its domestic component, consisting of domestic credit created by the monetary authorities and foreign components, that is, international reserves.

The money demand is defined as positively related to the nations' nominal national income and is assumed to be a stable function in the long run. On the other end of establishing equilibrium in the money market lies the money supply in an economy which is to a large extent given by the monetary authorities of the nation exogenously. The monetary equilibrium condition requires maintaining stability in the money market.

If the disequilibrium results from the inflow of foreign nations' currency it is equivalent to having a balance of payment surplus whereby inflow outturns the outflow of nations' currency abroad. Suppose the increased money demand is not satisfied by the monetary authorities through a rise in the domestic component of the money supply. In that case, the foreign exchange component shall increase creating an imbalance in the external account of the nation. Similarly, suppose the disequilibrium in the money market exists in the form of a higher money supply or a shortage in money demand. In that case, the foreign component of the monetary base flows out of the nation thereby creating a deficit in the nation's balance of payments account. Thus a surplus in the balance of payments accounts results from an excess in the stock of money demand, which is not satisfied by the rise in domestic components of the nation's monetary base, and BoP deficit

results from the excess in the stock of money supply which is corrected by the outflow of foreign nations' currency reserves from the domestic economy.

This is the scenario of a nation's monetary stance when it operates in a fixed exchange regime where its monetary authorities have no control over its money supply in the long run.

10.2.4.3 Monetary approach and flexible exchange rate

The monetary approach under a flexible exchange rate can be expressed as one, where currency depreciation results from an excess in the stock of money supply whereas currency appreciation results from an excess in the stock of money demanded in a given economy at a point of time. Since a higher stock of money supply is equivalent to having inflation in the domestic nation, it can be stated that a nation experiencing inflationary trends will experience depreciation in the foreign currency price of its domestic currency or will find its exchange rate rising. Alternatively, a nation that experiences lower inflation than other nations will experience an appreciation of its domestic currency or will find its exchange rate falling. Under a Flexible exchange rate, a benefit occurs to the nations in the form that a nation experiencing higher inflationary pressures will tend to transmit their inflation to the other nations through their rising imports rather than directly by exports of money reserves. Since alterations in the stock of imports and exports of goods and services is a real phenomenon that usually takes more time than any monetary change, the nations are somewhat shielded from the inflationary pressures as this rising import phenomenon is largely determined by structural conditions prevailing in any given nation.

Under a flexible exchange rate regime, nations' monetary authorities uphold dominant control over the stock of their money supply and can therefore implement monetary policies to suit their country's monetary environment. Under this, the external balance is immediately corrected by automated changes in the level of exchange rates and thus no inflow or outflow of domestic or foreign nations' currency assets is required to maintain equilibrium unlike in the fixed exchange rate regime.

Under both fixed and flexible exchange rates, the balance of payments surplus or deficit is adjusted through money market equilibrium. However, the equilibrium in both regimes is attained differently. Under fixed exchange rates, quantities of money adjust gradually through reserve flows to bring equality between actual and desired money stock. Under floating exchange rates, money market equilibrium occurs through domestic price changes that alter the valuation of the money stock.

The transition to floating exchange rate regimes in the 1970s changed the focus from the total BoP to its various subaccounts like the Current and Financial Account balances. These are the indicators of economic activities and currency repercussions to come. Although most nations strive for current account

surpluses, it is not clear that a balance on the current or the capital account, or a surplus on the current account, is either sustainable or desirable.

10.2.4.4 Criticisms of the monetary approach to BoP

This has been criticized on the following grounds:

- 1) Demand for money is not constant as assumed in the monetary approach to the balance of payments. In fact, in the real world, demand for money is only found to be stable in the long run and not in the short run.
- 2) A direct link is assumed between money supply and equilibrium in the balance of payments account. This indicates that the central bank has control over the inflow and outflow of foreign exchange reserves in the country in consonance with surplus or deficit in the balance of payments accounts. However, economists have argued that such sterilization operations' are not feasible in the present era of globalization where financial markets are highly integrated and countries are following an approach of flexible exchange rates.
- 3) The automatic self-adjustment mechanism highlighted in the monetary approach is feasible in long run. But it completely disregards the balance of payments disequilibria that exist in the short run. This is because the adjustments in the real sectors of the economy are not feasible to make in a short time. Thus, it will not be wrong to call the monetary approach a long-run approach to adjustments in the balance of payments.
- 4) The approach disregards factors other than monetary factors such as expansion or contraction of domestic credit that may create disequilibrium in the balance of payments account. It is asserted that completely disregarding the real sector disturbances is not a realistic assumption, and as such, disequilibrium may lead to adjustments through different routes in the balance of payments accounts.

The above flaws doubt its usefulness as a foundation for effective balance of payments management in developing countries. It is consequently argued that efforts to correct payments deficits should look beyond monetary factors and consider tackling external debt problems and ensuring resource flows into deficit countries.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Trace the evolution of the monetary approach to BoP.

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- 2) Given the assumptions of the monetary approach, how can you analyse that the BoP is a monetary phenomenon?

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- 3) Point out the criticisms against the monetary approach to BoP..

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- 4) a) Which of the following will prevail in the BoP account when money supply exceeds money demand?

- i) Deficit
- ii) Surplus
- iii) Neither deficit nor surplus.

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- b) What condition prevails in the BoP account when money demand exceeds the level of the money supply?

- i) Deficit
- ii) Surplus
- iii) Neither deficit nor surplus

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- 5) Explain how the monetary approach to BoP is essential:

- a) Short run phenomenon.
- b) Long run phenomenon.
- c) A phenomenon of short and long run both..

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10.2.5 Synthesising all the approaches to BoP

This section aims to provide a broad synoptic view of all the approaches. In the absorption approach income or expenditure clears the balance of payments. The monetary approach concentrates on the accumulation of money balances only. The stock-flow consideration of the monetary approach is the essential difference between the monetary approach and the elasticities and absorption approaches, where the latter two consider the flow equilibrium only. The monetary approach, like the absorption approach, stresses the need for reducing domestic expenditure relative to income, to eliminate a deficit in the balance of payments. However, whereas the absorption approach looks at the relationship between real output and expenditure on goods, the monetary approach concentrates on deficient or excess nominal demand for goods and securities, and the resulting accumulation or de-accumulation of money.

The different theories seek to explain different aspects of the balance of payments, so they are considerably complementary rather than competitive. The elasticities or traditional approach seeks to explain the balance of trade; the Keynesian theory explains the current balance; the monetary theory explains the overall balance. The theories are complementary in another sense. The traditional theory is a microeconomic theory of adjustment in the goods market; the *Keynesian* theory (a macroeconomics theory) in the goods market and the monetary theory (a theory of disequilibrium) in the money market. However, a complete analysis must take account of both real and financial factors so neither Keynesian nor monetary analyses can be complete in themselves. A complete analysis needs to look at both stocks and flows, *Keynesian* and monetary analyses are complementary as much as competitive. The monetary approach consists of the combination of two models: the money market and the foreign exchange market. The changes in the money market lead to changes in real returns on securities denominated in different currencies. Then investors' subsequent reaction to the changes in real returns leads to changes in the relevant exchange rate.

According to *Alan Barrett*, the monetary- approach to the balance of payments offers a richer and more satisfying analysis of the balance of payments as opposed to approaches which overshadowed it in the middle of the 20th century. The monetary approach to BoP analysis 'has been influenced directly by the changing character of international economic problems; in addition, however, it has also been affected by changing methodological fashions in the mainstream of economic thought'. The monetary approach to the balance of payments has drawn a good deal of attention from researchers.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Three major approaches to analysing the economic impact of currency depreciation are:

- a) the elasticities,
- b) the absorption, and
- c) the monetary.

Distinguish among the three.

.....

.....

.....

2) Are these approaches complementary? Give your answer.

.....

.....

10.3 LET US SUM UP

- Approaches to BoP are commonly known as the elasticities, absorption, and monetary approaches. The elasticity approach emphasizes the relative price effects of depreciation and suggests that depreciation works best when demand elasticities are high. As stated in the Marshall-Lerner condition, if the sum of the elasticities is less than one, currency depreciation will cause deterioration in a nation's trade position. A nation's trade balance remains unaffected by depreciation if the sum of the demand elasticities equals one.
- The absorption approach emphasizes the income effects of currency depreciation. According to this view, depreciation may initially stimulate a nation's exports and production of import-competing goods. The implication is that a decrease in domestic expenditure relative to income must occur for depreciation to promote trade equilibrium.
- The Keynesian analysis is an analysis of flows-flows of goods that are traded, and of income. The monetary analysis is an analysis of stocks, especially money stocks. The monetary approach stresses the effects depreciation has on the purchasing power of money and the resulting impact on domestic expenditure levels. The monetary approach establishes the link between BoP deficits, the intervention required in the foreign exchange market and the amount of money supply to prevail in a fixed exchange regime.

- The monetary approach considers inter-relationships among various markets and the inter-relationship between stock and flow equilibrium. The monetary approach focuses exclusively on investors' decisions between two securities. It considers investment-related factors in exchange rate determination. Investors compare the real interest rates in two countries.
- Under flexible exchange rates remember that the exchange rate bears all the strain of balance to payments imbalance. The main difference between the balance of payments deficit corrections under fixed and floating exchange rates is the absence of any need for expenditure reduction under floating rates.

10.4 KEY WORDS

The Asset Approach to BoP

Pays its exclusive focus on the behaviour of international investors who are trying to decide between securities denominated in two different currencies. Based on this, the monetary approach is also called the *asset approach to exchange rate determination* because by definition the price of one national money in terms of another.

Domestic credit policy

A way of measuring growth in the total amount of money in a country's economy at a particular time.

International reserves

International reserves are any kind of reserve funds that can be passed between the central banks of different countries. International reserves are an acceptable form of payment between these banks. International reserves are a country's '*external assets*'—including foreign currency deposits and bonds held by central banks and monetary authorities, gold and SDRs.

Sterilization policy

This is a form of monetary action in which a central bank seeks to limit the effect of inflows and outflows of capital on the money supply. It most frequently involves the purchase or sale of financial assets by a central bank and is designed to offset the effect of foreign exchange intervention.

10.5 SOME USEFUL REFERENCES

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10.6 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check your progress 1

- 1) See section 10.2.1
- 2) See section 10.2
- 3) See section 10.2.3
- 4) See sections 10.2.1, 10.2.3, and 10.2.3 and frame your answer.
- 5) See sections 10.2.1, 10.2.3, and 10.2.3 and make the differences clear.

Check your progress 2

- 1) See section 10.2.4
- 2) See section 10.2.4.1
- 3) See section 10.2.4.4
- 4) a) see section 10.2.4.1
b) see section 10.2.4.1
- 5) See section 10.2.4.1

Check your progress 3

- 1) See section 10.2.5
- 2) See section 10.2.5



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